Charm Industrial Isometric Standard Project Validation Report

Company / GHG Removal Supplier	Charm Industrial
Facility Name	MiniForge Facility Fort Lupton, CO, USA
Project	Charm Great Plains Kansas Bio-oil Geologic Storage
Production Facility Address	2360 S Rollie Ave, Fort Lupton, CO 80621
Greenhouse Gas Removals	NA – Validation Only
Removal Method	Bio-Oil Geological Storage
Validation Period	March 1, 2024 – February 28, 2029 (Update to original validation)
Verifier/Validation and Verification Body	Tim Hansen (Lead) Kyann Hadife (QA) 350Solutions, Inc.
Signature	Timothy A. Hansen Digitally signed by Timothy A. Hansen Date: 2025.08.04 11:50:43 -04'00'

ISSUED:

AUGUST 4, 2025

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ISOMETRIC STANDARD VALIDATION AND VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Kyann Hadife
Removal Method: Bio-oil Geological Storage		
Report Date: August 4, 2025		
Document No: 350VR-CHARM-ISO2405V2.0		
Rev: FINAL v2.0		

** primary contact/lead author*

1. INTRODUCTION

350Solutions was contracted to perform a project validation for additional project activities to be completed by Charm Industrial as part of their bio-oil geologic sequestration process for carbon removal (CDR). Charm's process was initially validated in July, 2024 for the use of third-party bio-oils and sequestration via injection into salt caverns [1]. Charm has since developed, installed, and initiated operation of its own biomass pyrolyzers at its MiniForge facility in Fort Lupton, Colorado, producing bio-oil which will then be treated and sequestered using the process identical to that which has been previously validated. This validation is solely for the Charm Fort Lupton MiniForge pyrolysis facility and ancillary activities and processes related to the use of the Charm-produced bio-oil as an injectant in the storage process.

The validation includes observation of operations during a site visit, collection of files and evidence related to the Charm pyrolysis process, review of data collection and handling procedures, and evaluation of adherence of the project, as summarized in the updated Charm Industrial Kansas Bio-oil Sequestration Project Design Document v. 1.7 (9/29/2024) [2], for adherence to the relevant Isometric Standards and Protocols.

350Solutions Inc. declares that we are an impartial verifying body, free from any conflicts of interest, capable, and qualified to complete this VVP according to the Isometric Standard and certified protocols utilizing a reasonable level of assurance. 350supported the independent development of the Isometric Bio-oil protocols as well as protocol modules for lifecycle emissions calculation (Energy, Transport, Embodied emissions).

350Solutions conducted a validation of the Charm bio-oil production process and associated greenhouse gas removal claims and reviewed administrative details to verify compliance with the requirements of the following standards, protocols and modules as required by Isometric:

- Isometric Standard v1.4.0 (August 12, 2024) [3]
- Bio-Oil Geological Storage Protocol v1.1 (September 13, 2024) [4]
- Biomass Feedstock Accounting Module v1.2.1 (May 20, 2024) [5]
- Biomass or Bio-oil Storage in Salt Caverns v1.1 (September 13, 2024) [6]
- Embodied Emissions Accounting v 1.0 [7]
- Energy Use Accounting v 1.1 [8]
- Transportation Emissions Accounting v 1.0 [9]

The validation and verification began with a site visit on August 30, 2024 at the Fort Lupton, CO MiniForge facility. Following the visit, a review of documents submitted by Charm prior to and following the site visit was conducted. Follow up questions and documentation requests discussed on site, documented by Charm in email on August 30, and modified by 350 on September 6. Responses from Charm were provided via additional file uploads and notes in an information request log throughout the remaining validation (August 30 through October 3, 2024).

A summary of the facility and validation is provided below.

TABLE 1. CHARM MINIFORGE VALIDATION SUMMARY

Validation Summary	
GHG Removal Supplier	Charm Industrial
Production Facility Name	MiniForge Charm Pyrolysis facility
Removal Method	Bio-Oil Geological Storage: Underground injection of bio-oil into salt caverns
Validation and Verification Report Date	October 4, 2024 (Original) August 4, 2025 (Revised)
Site Visit Date	August 30, 2024
Production Facility Location (Address and GPS Coordinates)	Bio-oil Production: Charm Industrial 2360 S Rollie Ave, Fort Lupton, CO 80621 40.076570, -104.804680 Biomass Supply: Altitude Forestry - McArthur Gulch Forest Restoration & Fuels Reduction Project Bailey, CO, 80421 Biomass Storage: ~13522 CO Rt 8, Fort Lupton, CO 40.040556,-104.797500 Bio-Oil Pre-Processing (previously validated): Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection (previously validated): Vaulted Deep 7513 KS-14, Hutchinson, KS 67501 37.966035, -97.947845
Verification Type & Standards	Isometric Standard Validation, including on-site visit and facility validation for additional activities operated by Charm Industrial (does not include validation of pre-processing and injection systems and facilities); Isometric Standard v1.4.0 (August 12, 2024) Bio-Oil Geological Storage Protocol v1.1 (September 13, 2024) Biomass Feedstock Accounting Module v1.2.1 (May 20, 2024)

	Biomass or Bio-oil Storage in Salt Caverns v1.1 (September 13, 2024) Embodied Emissions Accounting v 1.0 Energy Use Accounting v 1.1 Transportation Emissions Accounting v 1.0

2. TECHNOLOGY DESCRIPTION

Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. The facility has operated in an R+D capacity since December 2022 in order to develop Charm's internal pyrolysis technology. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of 4 pyrolyzers, with plans for up to 12-18 pyrolyzers total to be installed. The facility also continues to serve as a research and development and demonstration facility.

Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of the excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Charm uses woody biomass sourced from government-funded wildfire mitigation and restoration projects in Colorado. Currently, Charm is working with Altitude Forestry to source the non-merchantable logs removed from the forest as a part of this mitigation work. This woody biomass currently has no use and limited alternative disposal pathways.

Once sufficient bio-oil is produced at the Charm MiniForge facility, Charm contracts with transport providers to deliver bio-oil to Charm's pre-processing facility in El Dorado, KS. At this facility, the oil is introduced to storage tanks and a treatment system, which was constructed/commissioned in 2023-4. This site allows Charm to treat the bio-oil prior to injection to ensure that it is fully compliant with the requirements for underground injection, such as pH and chloride content, specified by the Kansas Department of Health and Environment (KDHE) and Vaulted's Class V injection permit. Following treatment, samples are taken for CHN, TAN, and Density testing and the oil is transported to the Vaulted injection facility.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, the oil is trucked to Vaulted's Great Plains Facility in Hutchinson, KS for injection into an underground salt cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored on site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are detailed in their own PDD and has been previously validated [1].

A summary of process inputs and outputs for the Charm Pyrolysis system is provided in Figure 1. A photo of the Charm pyrolyzer is provided in Figure 2.

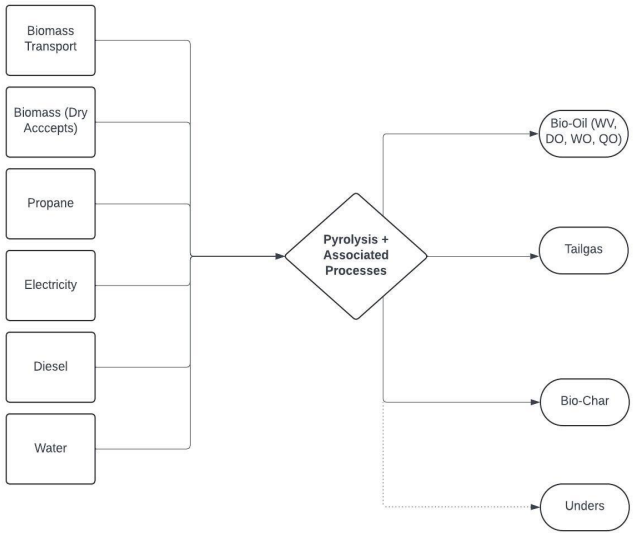

FIGURE 1. CHARM PYROLYSIS PROCESS INPUTS AND OUTPUTS SUMMARY

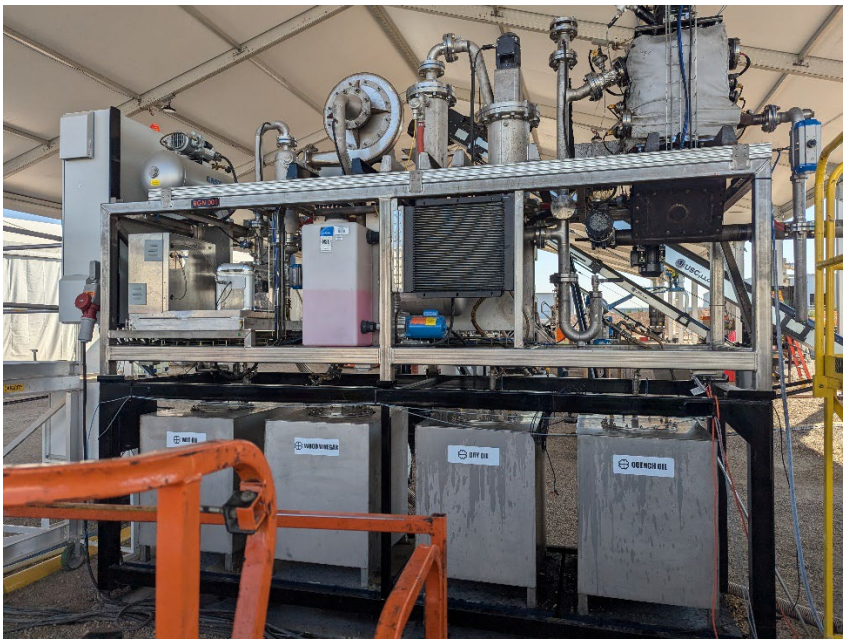

FIGURE 2. CHARM PYROLYZER AT FORT LUPTON MINIFORGE FACILITY

3. VALIDATION PROCESS SUMMARY

3.1. VALIDATION APPROACH

A planned series of validation activities were conducted by 350Solutions to independently validate and verify the production facility and its operations. The validation was conducted following the specifications of the Isometric Standard V1.4.0 and Bio-oil Geologic Sequestration Protocol v 1.1 as primary requirements, with numerous additional modules referenced (see Table 1. Charm MiniForge Validation Summary). Specific validation activities conducted are summarized in Table 2. A completed

Isometric Methodology Compliance Checklist used during the verification is attached to this report as Appendix 1. Photographs of the facility, equipment, and operations are provided in Appendix 2. The list of files that were reviewed are listed in Appendix 3. The follow-up request for information is documented in Appendix 4. Verifier qualifications are attached as Appendix 5.

TABLE 2. VALIDATION ACTIVITIES

Date(s)	Validation Activity	Validation Tasks	Documents Reviewed
August 30, 2024	Onsite Visit	- Opening meeting and process walk through - Witness of operations, measurement points, and instrumentation - Confirmation of company and facility administrative details - Confirmation of facility environmental and social safeguards - Review of raw material sources and sustainability - Review of system inputs and outputs - Review of MRV approach and measurements - Observation of biomass stockpile site	- Verifier observations of operations, measurement points, and instrumentation (see compliance checklist) - Walk through of PDD components and supporting documents found in Appendix 3 - Identification and request of additional supporting documentation as evidence to claims in PDD
September 3 – 6, 2024	Initial Document Review	-Detailed review of PDD, supporting documentation, data and measurement summaries, biomass feedstock sourcing, environmental, safety, and social safeguards supporting documentation	- Complete file list provided in Appendix 3
August 30 and September 6, 2024	Follow-up Request for Information	Issued request for follow-up information and supporting evidence. Information request list documented by Charm on August 30 and reviewed and commented / updated by 350 on Sept. 6.	- See Appendix 4
September 30 – October 3, 2024	Validation completion and report development	Reviewed final validation evidence and documentation. Completed 350 Validation Checklist for Isometric Bio-Oil Sequestration Projects. Developed draft Validation Report	- See Appendix 1 for completed checklist
October 4, 2024	Submitted Validation Report Draft	Submitted draft validation report to Isometric and Charm for review	

3.2. PROCESS INPUTS & OUTPUTS

The charm biomass pyrolysis process uses chipped waste timber from forest management operations as its primary feedstock. Under normal operations the mixed woody waste, which is maintained in a stockpile near the pyrolysis site, is at a low moisture content, is chipped on site, and loaded into hopper wagons for transfer to pyrolysis. Wood chips are weighed and sampled at the pyrolysis facility and transferred into 'pro-boxes' which are tracked and weighed upon entry into pyrolysis operations. Charm utilizes waste heat generated by the pyrolysis process to further dry wood chips prior to feeding into the pyrolysis reactor.

A summary of all process inputs and outputs relevant to the Charm pyrolysis process is provided in Figure 1 and in Table 3 below. 350 has validated that all observed inputs and outputs are accounted for and are addressed in the GHG accounting procedure and monitoring plans presented by Charm.

TABLE 3. CHARM MINIFORGE PROCESS INPUTS AND OUTPUTS*

Input / Output	Process / Facility	Input or Output?	Included in GHG Accounting	Notes
Biomass	Pyrolysis	Input	Yes	Primary feedstock
Water	Pyrolysis	Input	Yes	Primarily for quench
Diesel	Pyrolysis	Input	Yes	For material handling equipment
Electricity	Pyrolysis	Input	Yes	For pyrolysis equipment operation and onsite support activities
Propane	Pyrolysis	Input	Yes	Used for flare support and pyrolyzer startup (torch in)
Equipment (all pyrolyzer, storage, facility, site transport and material handling, and similar equipment)	Pyrolysis	Input	Yes	All equipment is accounted for in inventory. Consumables and spare parts are also tracked.
Bio-oil (combined dry oil, wet oil, quench oil, wood vinegar)	Pyrolysis	Output	Yes	Bio-oil fractions are stored separately, but are combined when transferred to pre-processing site
Tailgas	Pyrolysis	Output	Yes	Tailgas is destroyed in a flare
Biochar	Pyrolysis	Output	Yes	Biochar is currently collected and stored for future sale or use
Unders (biomass waste)	Pyrolysis	Output	Yes	Unders are biomass feedstock (chips) that are below the preferred size specification. These are collected and currently disposed of
Wastes	Pyrolysis	Output	Yes	Includes solid waste, liquid waste, hazardous waste resulting from pyrolysis operations, including use and disposal of cleaning and maintenance products
Diesel	Biomass Transport	Input	Yes	Fuel usage for truck delivery of biomass
Diesel	Bio-Oil Transport	Input	Yes	Fuel usage for truck or rail delivery of bio-oil

**Note: all other process inputs and outputs for bio-oil pre-processing and bio-oil storage (injection) are accounted for in prior validation and not addressed here as no changes have been made in those processes.*

4. VALIDATION SUMMARY

4.1. SUMMARY OF VALIDATION FINDINGS & VALIDATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the Charm Project Design Document and supporting Validation information and during the site visit and follow up:

☐ ***Meets the requirements of the Isometric Standard V1.0.0 and Biomass Geological Storage Protocol for Validation***

☒ ***Meets the requirements of the Isometric Standard V1.0.0 and Biomass Geological Storage Protocol for Validation with minor modifications required***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.0.0 and Biomass Geological Storage Protocol for Validation***

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance, collected data, and supporting documentation and has found that the data presented in the Project Design Document meet the requirements of the Isometric Standard and Bio-oil Geological Storage protocol and related modules, with minor modifications required to ensure the planned GHG Accounting and Monitoring Plans are implemented properly for the first reporting period and verification.

Our opinion is provided with a reasonable level of assurance for activities at the Charm MiniForge pyrolysis facility and associated activities (biomass harvesting, transport, and storage, and bio-oil transportation).

Notice: 350Solutions, Inc. declares that we are an impartial auditor, free from any conflicts of interest, capable, and qualified to complete this audit according to the Isometric Standard and related Validation and Verification Body Requirements. Verifications and audits conducted by 350Solutions are based on an evaluation of technology performance and CO_2 removal claims via site visit observations and review of data submitted by the audited company. Audits are completed in accordance with rules and methodologies specified by Isometric and utilizing the appropriate quality assurance procedures established under the 350Solutions accredited ISO 17020/14034 Quality Management Program, noting that this verification is not a fully compliant ISO 14034 verification. 350Solutions makes no expressed or implied warranties as to the performance of the technology and does not certify that a technology will always operate at the levels verified, nor that it meets all state, local, or federal legal requirements.

4.2. SUMMARY OF VALIDATION FINDINGS

A summary of findings and any identified issues associated with each requirement of the Isometric Standard V1.0.0, Biomass Feedstock Accounting Module, and Biomass Geological Storage Protocol are summarized below. Please see Appendix 1 for a detailed description of all requirements and the validator's assessment and findings.

TABLE 5. VALIDATION FINDINGS

Standard / Protocol Section Reference	Validation Topic	Findings
2.1.	*Validated against approved Protocols*	Acceptable with modification – The following Isometric Approved Protocols were used: Biomass Feedstock Accounting Module, Bio-oil Geological Storage Protocol and referenced modules and referenced modules. Project Proponent must provide updated GHG Statement, uncertainty analysis, and bio-oil sampling and analysis results for Charm-produced bio-oils when data is available and prior to first verification of removals via injection of Charm-produced bio-oil.
2.5.1 / 7.1	*Boundaries*	Acceptable – Project Boundary and GHG accounting boundary are defined geographically at the point of transportation of the biomass until it is emplaced at the Great Plains Facility in Hutchinson, KS and incorporate all embodied (cradle-to-grave) and monitoring GHG emissions. This validation is for additional activities to be included during the crediting period from March 1, 2024 through February 28, 2029 (5 year validation period)
2.5.2 / 7.2	*Baseline*	Acceptable – Biomass qualifies as EC5, EC8, EC14 and C1 for direct market leakage, indirect market leakage, baseline scenario, and counterfactual accounting under the Isometric Biomass Feedstock Accounting Module Requirements. Biomass utilized is from a government forest management and fire reduction program and has no other current use and limited disposal options.
2.5.3	*Additionality - Financial*	Acceptable – It has been stated that no other income streams are currently obtained. Project pays a fee for biomass feedstock.
2.5.3	*Additionality - Environmental*	Acceptable with modification – Project is projected to be net negative relative to the counterfactual case. Final GHG statement must be provided prior to or as part of first verification of removals using Charm bio-oil to ensure net negativity based on actual production data. Charm must also provide confirmation of the determination of the final bio-oil product as non-hazardous. Proper sampling and monitoring plans are in place but have not been completed due to limited bio-oil production at the time of validation.
2.5.3	*Additionality - Regulatory*	Acceptable – There are no regulatory requirements calling for disposal of biomass wastes in salt caverns in the local, state, nor national jurisdictions.
2.5.4	*Leakage*	Acceptable – feedstock qualifies as EC5 and EC8 for direct and indirect market leakage impacts – indicating no leakage is considered as feedstock is a waste product with no current use.

		Any potential leakage due to release of GHGs via vaporization or decay of from on-site storage of biomass or bio-oil is limited as biomass counterfactual would be storage and decay and bio-oil is injected the same day as delivered, reducing any significant release potential. Below ground reversal and leakage is monitored using a gas sensor and deducted from totals if it occurs.
2.5.5	*Default Emission Factors, Proxies, and Models*	Acceptable – Carbon content measurement is currently completed for every batch of bio-oil injected. Conservative values for items such as transportation fuel efficiency, equipment sizes for embodied carbon calculation are used. Default emission factors citations include literature citations and are acceptable.
2.5.6	*Common Calculation Factors*	Acceptable – All non-metric values are converted to the equivalent metric basis. All emission factors are properly cited.
2.5.7	*Uncertainty*	Acceptable with modification – Uncertainty and sensitivity analyses provided for prior validation and similar description provided for Charm bio-oil production. Project proponent must provide updated uncertainty and sensitivity analyses based on actual data from Charm bio-oil production process prior to first delivery of removals and subsequent verification using Charm bio-oil to confirm uncertainty analysis approach.
2.5.8	*Durability and Monitoring*	Acceptable – Peer reviewed literature indicating durable removal of >1000 years provided. No change to durability monitoring from prior validation.
2.5.9	*Risk of Reversal*	Acceptable – Risk Assessment Questionnaire score was 0 which indicates a Very Low risk. No change to durability monitoring from prior validation.
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	Acceptable – The Charm pyrolysis site (MiniForge) has all required operations and environmental permits, safety plans and staff training. Charm has become active in the Fort Lupton community and plan to host additional community engagement meetings in late 2024 to improve community engagement and response to any concerns.
2.5.11 / 7.3	*Life Cycle Assessment (LCA) Policies*	Acceptable – the proposed GHG accounting methods align with the protocol and relevant modules. A draft or final GHG statement must be provided prior to or as part of first verification of removals using Charm bio-oil to ensure full implementation of accounting practices is confirmed.
3.1	*Ownership*	Acceptable – Feedstock acquisition contract states transfer of rights to “monetization of environmental benefits and attributes”.
3.2	*Documentation*	Acceptable – PPD completed appropriately and submitted.
3.3	*Eligibility*	Acceptable – Project registered with Isometric.
3.4	*Project Crediting*	Acceptable – Project period is complete, and all removals have been verified.
3.5	*Stakeholder Input Process*	Acceptable - The project hosted a community engagement meeting prior to opening with no objections, there were queries on job creation of which there were a few jobs openings created. Ongoing

		input is primarily managed through the KDHE permit reporting and feedback process.
3.6	*Regulatory Compliance*	Acceptable – Maintenance of well operation permits would meet requirements.

4.3. VALIDATION ISSUES

The validation of the additional activities associated with Charm's production of bio-oil at the MiniForge facility has not revealed any major concerns, non-compliance, or material misstatements. However, because of the state of operations of the charm pyrolysis facility, Charm-produced bio-oil has not yet been injected and all data required to complete a GHG Statement and qualification of Charm bio-oil was not available for this validation. Therefore, there are minor concerns which must be addressed prior to Charm-produced bio-oil injection for crediting purposes. These are summarized below in the following critical Forward Action Requests, which should be considered as requirements prior to verification of removals using the Charm Bio-oil:

TABLE 4. CRITICAL FORWARD ACTION REQUESTS - CHARM MINIFORGE

ID	Type	Finding / Issue	Audit Evidence	Protocol Reference	Required Action	Conclusion / Resolution
1	Forward Action Request	Example LCA not provided demonstrating net negative LCA. Charm PDD v1.7 provides full LCA calculation description and monitoring plan, providing adequate documentation of approach for LCA and data use, but full LCA not provided due to status of project development (lack of injections using Charm bio-oil)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Bio-oil Geological Storage 4.0	Provide initial LCA for Charm-produced bio-oil demonstrating net negative emissions and documenting LCA data usage as required by protocol	To be confirmed at Verification
2	Forward Action Request	Uncertainty analysis for Charm Bio-oil production is not provided. Charm PDD v1.7 provides uncertainty analysis and sensitivity analysis description, providing adequate documentation of approach, but full analyses not provided due to status of project development (lack of injections using Charm bio-oil). Charm has demonstrated full uncertainty and sensitivity analyses in prior verification/validation using third-party bio-oil, providing evidence of proper completion	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Standard - 2.5.7 & Appendix A; Biomass Geological Storage - 6.5	Provide uncertainty and sensitivity analysis for GHG Statement for production and utilization of Charm bio-oil	To be confirmed at Verification
3	Forward Action Request	Analysis of bio-oil for hazardous determination (TCLP or similar analysis as required under protocol) has not been completed due to lack of bio-oil production for injections		Bio-oil Geological Storage 5.0	Provide analysis in compliance with the requirements of protocol (Bio-oil testing results using US EPA RCRA SW0846 test methods) to demonstrate non-hazardous claim	To be confirmed at Verification

4	Immaterial Non-conformity	Charm biomass storage site location information is not included in the PDD	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf (SectionA)	Bio-oil Geological Storage 6.1	Modify PDD to include biomass storage site in Ft. Lupton operated by Charm	PDD modified. Resolved.
5	Immaterial Nonconformity	Crediting period indicated in PDD as '2024/12/03'. Verification period for prior removals was March 1, 2024.	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf		Modify PDD to reflect proper Crediting period starting March 1, 2024 and continuing for 5 years	PDD modified. Resolved.

In general, Charm has complied with all requirements of the Isometric Standard and the Bio-oil Geological Sequestration Protocol. All requirements, aside from those mentioned above, have been fully addressed in the PDD and were observed to be implemented for the Charm MiniForge bio-oil production facility and related activities.

4.4. RECOMMENDATIONS & OPPORTUNITIES FOR IMPROVEMENT

Based on the above VVP findings and issues, as well as on-site observations, 350Solutions has the following recommendations for improvements prior to the next verification.

Forward action requests (FAR) are requests for future verifiers to review and evaluate during future verifications involving the production and injection of Charm-produced bio-oil. These are recommendations based on information obtained during the validation that indicate changes in system operations may occur or additional information may be needed which should be confirmed.

Recommendations are those items that may be addressed by the project proponent for improving the quality of data, accuracy, and verifiability of the GHG Removal claims in the future.

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS FOR IMPROVEMENTS

ID	Type	Finding / Issue	Audit Evidence	Protocol Reference	Requested Action	Notes
6	Forward Action Request	Because Charm-produced bio-oil injections have not begun as of writing, bio-oil sampling and analysis has not been completed in accordance with the Monitoring Plan and requirements of the protocol	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Bio-oil Geological storage 7.4.1.1.1, Biomass or Bio-oil Storage in Salt Caverns 3.1.1	Verify bio-oil sampling completed in accordance with requirements of Monitoring Plans as well as UIC permit in first verification of Charm bio-oil injection	Note that monitoring plan is adequate and compliant, and compliance by Charm has been demonstrated in prior verification/validation using third-party bio-oil, so capability is demonstrated.
7	Forward Action Request	Example LCA not provided demonstrating appropriate use of data, emission factors, and calculation methods outlined in PDD	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Bio-oil Geological Storage 7.X	Demonstrate all aspects of LCA related to Charm-produced bio-oil and have verified	Charm PDD v1.7 provides full LCA calculation description and monitoring plan, providing adequate documentation of approach for LCA and data use, but application in full LCA not confirmed due to status of project development

8	Forward Action Request	Air permitting requirements are documented and currently complete. However, with addition of pyrolyzers and increased operation, air permitting notice and confirmation must be provided to CO.	Charm Industrial Wood Chip Pyrolyzer Burner APEN Cover Letter (1).pdf Air Permitting Pathway in CO.docx	Bio-oil Geological storage 5.0	Monitor requirement for submitting air permitting notification (APEN) in future verifications	Air permitting is currently not required, but APEN notice will be required as operations scale
9	Forward Action Request	Site operations are increasing and scaling. As changes occur, ensure permitting and compliance requirements are met		Bio-oil Geological storage 5.0	Review and validate permitting requirements and compliance status at future verifications	Currently in compliance
10	Forward Action Request	Claimed improvements in community engagement activities are planned, including open community meeting (November, 2024) and follow-up open house (December 2024)	Colorado Outreach and Engagement 2024.pdf	Standard - 3.5	Update community engagement records to document planned activity. Include in future verifications	
11	Forward Action Request	Logs of spills, leaks, and process upsets are minimal due to lack of current operating time.		Bio-oil Geologic Storage 7.4.1.3	Provide updated logs for first verification of Charm-produced bio-oil	
12	Forward Action Request	Several process changes are occurring in Q3/Q4 2024 that may impact LCA calculations. These include: - installation of additional pyrolyzers - replacement of flare with low-swirl burner			Review site operations and document all changes from operations observed during site visit. Verify changes are accounted for in LCA, including additional equipment inventory and embodied emissions (equipment and construction)	
13	Forward Action Request	Charm is installing a new burner to manage tail gas from pyrolysis systems. A new emissions test will be conducted following installation and commissioning	Verbal discussions during site visit		Provide new emissions testing report and methane emissions calculation documentation. Account for new tailgas emissions values in LCA and have verified in first verification for Charm bio-oil after changeover to low-swirl burner	
14	Forward Action Request	Because Charm-produced bio-oil injections have not begun as of writing, bio-oil sampling and analysis has not been completed in accordance with the Monitoring Plan and requirements of the protocol and UIC permit for injectate qualification and monitoring			Provide all required monitoring records for bio-oil analysis prior to verification using Charm Bio-oil	
15	Forward Action Request	Charm injection site (Vaulted Deep operated) is installing new gas monitoring equipment to better identify gas production at the wellhead / brine holding tanks. Updated information should be provided.			Provide specifications, design, installation information, calibrations, and operational data for the new gas monitoring equipment	

16	Recommendation	Evidence of reasonable community engagement via community support activities is provided. However, limited open community outreach and engagement has occurred, such as open town hall meetings. These are currently planned for November / December 2024, and such activities are highly recommended.	Colorado Outreach and Engagement 2024.pdf Verbal discussions during site visit	Standard - 3.5	Update community engagement records to document planned activity. Include in future verifications	
17	Recommendation	Changes in future operations may result in modifications to additionality assessment. Charm has indicated some potential for collecting tipping fees for certain feedstocks and the potential for sale of biochar as a co-product.	Verbal discussions during site visit	Standard - 2.5.3; Bio-oil Geological Storage - 6.4	None. Monitor operations for changes that may impact additionality and re-assess when required	
18	Recommendation	Transportation emissions utilize "the average "deadhead" or unloaded journey distance for a vehicle of this type according to USDA analysis". This is sufficient for accounting for return trips, but it may be preferable or more appropriate to use return trip to logging site if the trip route is known.		Bio-oil Geologic Storage 7.4.3 (Eq7)	Consider requesting return trip information from biomass delivery and bio-oil delivery transportation services and account for return trip accurately if appropriate	

5. REVISION HISTORY

Original date of issue: October 4, 2024

Version	Date Issued	Noted Changes
Draft Versions (v1.0)	October 4, 2024	NA
Draft Versions (v1.1)	October 8, 2024	Internal Review - Minor edits and clarifications
Draft Versions (v1.2)	October 8, 2024	External Review - Minor edits and clarifications
Final Version (v1.3)	October 18, 1024	Minor edits and clarifications
Final Version – Revision (v2.0)	August 4, 2025	Corrected Crediting Period dates to March 1, 2024 – February 28, 2029

References

[1] Futurepast, "Report on Futurepast's Engagement with Isometric Relative to its Charm Industrial Great Plains Bio-oil Sequestration Project," 2024.

[2] Charm Industrial, "Charm Industrial Kansas Bio-Oil Sequestration Project Design Document v 1.7," 2024.

[3] Isometric, "Isometric Standard v 1.4.0," 2024.

[4] Isometric, "Bio-oil Geological Storage," 2024.

[5] Isometric, "Biomass Feedstock Accounting," 2024.

[6] Isometric, "Biomass or Bio-oil Storage in Salt Caverns," 2024.

[7] Isometric, "Embodied Emissions Accounting," 2023.

[8] Isometric, "Energy Use Accounting," 2024.

[9] "Transportaion Emissions Accounting," 2023.

See Appendix 3 and Table 2 for list of specific files reviewed during the Validation.

APPENDIX 1: ISOMETRIC STANDARD & BIO-OIL GEOLOGICAL STORAGE PROTOCOL VALIDATION CHECKLIST

General Project Requirements

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Conservative assumptions	Choosing input parameters that will purposefully result in a lower net CO2 Removal than if using the median input values	Yes	Baselines	Shared distribution of data and method for making assumptions and decisions	Standard - 2.5.2; Biomass Geological Storage - 7.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Documentation of GHG calculations demonstrates conservative calculations for C content of injected bio-oil and metered data for major components of GHG emission calculations	Charm uses conservative assumptions in calculations which lead to an underestimate of total removals
Project Documentation	Project is thoroughly documented via a PDD which includes all required information including specific to bio-oil: - location information for biomass production, conversion, injection, and geologic storage formation - conditions of biomass use prior to project initiation - details on technologies, products, services related to biomass conversion, including production rates and volumes	Yes	Eligibility	Project Design Document (PDD)	Bio-oil Geological storage 6.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	All required PDD information is included in v1.7 PDD. Additional details on Charm Bio-oil production technology (pyrolyzer) could be added for the benefit of VVB and public, such as a process flow diagram, pyrolyzer technology description, and site description regarding numbers and operating schedules of pyrolyzers. Feedstock storage and pre-processing location must be added to PDD

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Site Location	Site is located in the US	yes	Eligibility	Project site address (injection / storage site)	Bio-oil Geological storage 4.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Visual confirmation of site locations	Confirmed
Bio-oil not used for EOR/HER	Bio-oil must not be used for enhanced oil/hydrocarbon recovery	Yes	Eligibility	Documentation of no oil / gas extraction from wells at the project site	Bio-oil Geological storage 4.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation
Net Negative LCA	Project provides a net negative CO2e impact (i.e. a CO2e removal) based on GHG statement	Yes	Eligibility	LCA completed in compliance with protocol Section 7	Bio-oil Geological storage 4.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations.	GHG Statement and Calculation details meet requirements for LCA calculations and indicate likely Net negative LCA. Full LCA is not provided as this is for a validation update with LCA to be provided at time of future verification
No emissions reductions or avoidance	Emissions reductions are simply lowering or avoiding future GHG releases from a specific entity or project	Yes	Baselines	Only Carbon Removal is acceptable - all positive emissions must be subtracted from Removals	Standard - 2.5.2; Biooil Geological Storage - 7.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations.	GHG calculation description for Charm Bio-oil sequestration does not account for any avoided or reduced emissions that should be excluded
Financial Additionality	An evaluation of the likelihood that an intervention that causes a climate benefit above and beyond what would have happened in a no-intervention Baseline scenario was the result of revenues from carbon finance	Yes	Additionality	Removals are the main purpose and only source of revenue of the Project; OR demonstrating that economic barriers would prevent Project implementation in the absence of Carbon Finance; OR in the absence of carbon finance, Removals only in excess of this historical baseline may be considered financially additional	Standard - 2.5.3; Bio-oil Geological Storage - 6.4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed. Charm is not recovering tipping fees from feedstock suppliers, is no selling co-products and is relying exclusively on carbon removal crediting for finance. Note: biochar sales are a potential future consideration as are feedstock tipping fees which should be monitored and reviewed for changes at future verifications

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Environmental Additionality	An evaluation of the likelihood that an intervention causes a climate benefit above and beyond what would have happened in a no-intervention Baseline scenario	Yes	Additionality	Net negative climate impact when compared to the Counterfactual scenario	Standard - 2.5.3; Bio-oil Geological Storage - 6.4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Counterfactual use of biomass and BAU case specifies feedstock is typically landfilled or disposed and has no current use. Therefore baseline scenario is full decay and re-emission of all biomass carbon. Project provides defined climate benefit based on net negative LCA (to be confirmed for each verification)
Regulatory additionality	An assessment that determines whether the Project activity is not required by an existing regulation or policy and therefore it goes above and beyond what would have happened in a no-intervention Baseline scenario	Yes	Additionality	The Project is not required by existing laws, regulations, policies, or other binding obligations, OR if the Project is legally required, the Project provides Removals that exceed the legal requirement	Standard - 2.5.3; Bio-oil Geological Storage - 6.4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Disposal of the forestry biomass from fire management via use pf pyrolysis and bio-oil injection are not required by any regulatory authority.
Biomass Feedstock Eligibility	Feedstock eligibility with respect to sustainability, leakage, and counterfactual scenarios; Agricultural or forestry residues only; Biomass is sustainably sourced	Yes	Eligibility	Documentation that demonstrates biomass feedstock eligibility, sourcing, counterfactual emission determinations Documentation of biomass type (purchase records, etc.) sustainable sourcing documentation	Bio-oil Geological Storage - 4.0; Biomass Feedstock Accounting - 2.1, 2.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Biomass utilized is provided by Altitude Forestry via a contract specifying use of fire management and related waste timber that has no othercurrent use and is contracted by the CSFS for removal from the Macarthur Gulch Forest Restoration and Fuels Reduction project. Biomass qualifies as EC5 for direct market leakage, EC8 for indirect market leakage, EC14 for counterfactual, and C1 (counterfactual is=0).

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Leakage	The increase or decrease in GHG emissions outside the geographic or temporal boundary of a project that results from that project's activities	Yes	Leakage	Shared assessment of potential increases in GHG emissions outside the defined project boundary, AND if there is leakage the Project outlines how it will be quantified and deducted from Removals	Standard - 2.5.4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	No leakage occurs as a result of this project. Biomass qualifies as EC5 for direct market leakage, EC8 for indirect market leakage, EC14 for counterfactual, and C1 (counterfactual is=0).
Conversion Process applicability	Biomass converted to bio-oil via pyrolysis or similar process or utilizxes bio-oil produced by a third party	Yes	Eligibility	Documentation / visual observation of process design and operation confirming pyrolysis	Bio-oil Geologic storage 4.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Visual confirmation of process in operation (Charm pyrolysis)	Confirmed
Bio-oil C Content	Bio-oil analysis is completed for every injection batch OR Initial sampling of 30 production batches, then sampling of at least every tenth batch minimum of 3 samples for each production batch Carbon content calculated for those that are not sampled in accordance with protocol (Eq 5, 6). Eligible samples are those from prior 6 months before any given production batch date	Yes	Monnitoring, LCA	Bio-oil testing results (See monitoring tab for requirements)	Bio-oil Geoligic storage 7.4.1.1.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	Confirmed - no change from previous Validation. Bio-oil content analysis has not been initiated for Charm produced bio-oil as it has not been deliveredto pre-pocessing or injection site for injection. Confirm at future verification
Reputability of sources	A source that would be widely considered trustworthy based on the process undertaken (e.g., peer review) or origin of the information (e.g., government body)	Yes	Default Factors, LCA, and Models	Proper citation of all sources, models, data, etc.	Standard - 2.5.5	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed via review of PDD. Confirm all are used properly in future verifications (full LCA calculations not available).
Data transparency	Open-ness about any non business-sensitive information that would be necessary in scientific replication of results	Yes	Default Factors, LCA, and Models	All models, and any modifications to them, their input parameters, or data used and validation results are clearly outlined	Standard - 2.5.5	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
GWP	Global Warming Potential (relative to CO2, which has a value of 1)	Yes	Default Factors, LCA, and Models	Global Warming Potential (GWP) calculations use 100-yr GWP factors, based on the current volume of the IPCC Assessment Report	Standard - 2.5.6	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Confirmed Confirm all are used properly in future verifications (full LCA calculations not available).
Consequential analysis	The analysis of specific Uncertainties, hazards and scenarios inherent in complex systems such as the natural and engineered environment, aiming to describe how systems-level environmentally relevant flows will change in response to possible decisions	Yes	Default Factors, LCA, and Models	The Project uses Consequential Analysis for its Life Cycle Assessment, OR the Project appropriately justifies the use of Attributional Analysis (e.g. on grounds of accuracy and feasibility)	Standard - 2.5.11	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation.
Uncertainty	Lack of knowledge of the exact amount of CO2 removed by a particular process, Uncertainty may be quantified using probability distributions, confidence intervals, or variance estimates	Yes	Uncertainty	The Project outlines a procedure for incorporating Uncertainties into a Conservative estimate of Removals via a Conservative estimate of input parameters, variance propagation, OR Monte Carlo Simulations	Standard - 2.5.7 & Appendix A; Biomass Geological Storage - 6.5	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Partially confirmed - no change from previous Validation for general approach. However, uncertainty analysis for Charm-pyrolysis bio-oils has not been provided and must be updated and confirmed at future verifications.
Verification Materiality	Materiaility threshold for differences between verified CO2 removals and GHG statement is less than 5%	Yes	Materiality	GHG Statement and verifier determination of GHG statement errors and omissions	Bio-oil Geoligic storage 6.1	Not applicable - validation only	Not applicable -to be confirmed at future verifications

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Durability (>1000 years required)	The amount of time carbon removed from the atmosphere by an intervention – for example, a CDR project – is expected to reside in a given Reservoir, taking into account both physical risks and socioeconomic constructs (such as contracts) to protect the Reservoir in question	Yes	Durability and Monitoring	The Project demonstrates a Durability of at least 1,000 years via containment mechanisms, where the conservatively estimated engineering and/or scientific methods for containment exceed 1,000 years OR via scientifically falsifiable hypotheses that can be used to show there is no alternative destination for carbon storage other than the Reservoir in question	Standard - 2.5.8	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation.
Reversal risk	The escape of CO_2 to the atmosphere after it has been stored, and after a Credit has been Issued	Yes	Durability and Monitoring	The Project has undertaken a full risk assessment to identify all possible mechanisms that may lead to Reversals and subsequent decreases in Durability	Standard - 2.5.8	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation.
Monitoring	Tracking a project in order to maintain awareness of changes and impacts over a certain period	Yes	Durability and Monitoring	There is a sufficient Monitoring Plan in place to quantify the amount of potential Reversal that may occur via each identified Reversal mechanism	Standard - 2.5.8	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation. Revised monitoring plan submitted in PDD that accounts for all required monitoring for the Charm bio-oil process
Buffer Pool	A common and recognized insurance mechanism among Registries allowing Credits to be set aside (in this case by Isometric) to compensate for Reversals which may occur in the future.	Yes	Durability and Monitoring	The Project has categorized and justified an assessment of Risk of Reversal, following the guidance in the Risk of Reversal risk questionnaire, to determine the appropriate Buffer Pool Size	Standard - 2.5.8 & Appendix B	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation. Low risk of reversal - utilizes minimum 2% buffer pool

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Socioeconomic harm	Socioeconomic harm	Yes	Environmental and Socioeconomic Impacts	The Project has demonstrated that it will do no net environmental, social or economic harm AND the PDD includes a summary environmental impact assessment (if required) documented risk assessment and mitigation plan for safeguarding working conditions, including safe handling and transport of bio-oil	Standard - 2.5.10; 3.2; 3.7; Bio-oil Geological Storage - 6.1; 5.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx EHS-0006 Emergency Action Plan - MiniForge.pdf EPA ID.pdf EHS-0011 Chemical Hygiene Plan (CHP) & Hazard Communication Program (HCS) - Google Docs.pdf Monthly Site Inspections -Template.pdf EHS Training Modules - MiniForge - Sheet1.pdf ADP MyLearning for MRV - Operators.pdf Charm Bio-Oil SDS.pdf Monthly Site Inspections -Template.pdf EHS Training Modules - MiniForge - Sheet1.pdf ADP MyLearning for MRV - Operators.pdf Charm Bio-Oil SDS.pdf SOP-0011 Bighorn Standard Operating Procedure_REV-B (1).pdf SOP-0007 Biomass Screening System Standard Operating Procedures.pdf SOP-0001 Peanut Wagon.pdf MiniForge IBC Tote Bio-oil Sampling SOP.pdf WI-0901 Bio-Oil Transfer.pdf SOP-0001 Peanut Wagon.pdf MiniForge IBC Tote Bio-oil Sampling SOP.pdf WI-0901 Bio-Oil Transfer.pdf Charm Industrial_Wood Chip Pyrolyzer Burner APEN Cover Letter (1).pdf MiniForge Compensation Summary.docx Reportable Safety Incidents TD.png	Confirmed. Additional documents provided regarding Charm pyrolysis process operation, EH&S requirements, site safety management, staff training. Documentation indicates strong effort to ensure safety and minimal impacts to staff, community, and environment

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Environmental harm	Environmental harm	Yes	Environmental and Socioeconomic Impacts	The Project has demonstrated that it will do no net environmental, social or economic harm AND the PDD includes a summary environmental impact assessment (if required)	Standard - 2.5.10; 3.2; 3.7; Bio-oil Geological Storage - 6.1; 5.0		
Environmental Permit compliance	All required permits for site operation are obtained and maintained by project proponent	yes	Environmental and Socioeconomic Impacts	documentation of activites on site and permit requirements assessment as well as copies of all required permits	Bio-oil Geoligic storage 5.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Charm Industrial_Wood Chip Pyrolyzer Burner APEN Cover Letter (1).pdf Copy of Emissions Performance Test Report EM-24-1203-001 REV 0.pdf Air Permitting Pathway in CO.docx Permit_24FTL-PW00057 - Ponds and Utilities	Confirmed. Charm has completed proper air permitting analysis and does not require permitting for up to 7 pyrolizers in operation. For up to 18 pyrolizers planned, Charm has prepared required notification to CO but will still not require a full air permit - just notification. Monitor APEN submission for air emissions in future verifications Charm has worked to develop proper stormwater plans and permit and has a permitted detention pond on site
Site Access and Operating permits	Project proponent has obtained required drilling permits, access agreements, encroachment permits, water, or land use permits	Yes	Environmental and Socioeconomic Impacts	documentation of activites on site and permit or access agreement requirements assessment as well as copies of all required permits	Bio-oil Geoligic storage 5.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation for injection and pre-processing sites. Confirm compliance with permits for new bio-oil in future verifications
Bio-oil non-hazardous documentation	Document whether bio-oil is hazardous or non-hazardous	yes	Environmental and Socioeconomic Impacts	Bio-oil testing results using US EPA RCRA SW0846 test methods	Bio-oil Geoligic storage 5.0	Results of bio-oil testing not complete. To be provided	Non-compliant. To be validated prior to first injection of Charm produced bio-oil
Post-cessation plan	Plan for closing a specific site that aligns with local regulations and requirements	Yes	Environmental and Socioeconomic Impacts	The Project has defined a Closure Plan that outlines post-cessation actions that are needed by the Project Proponent	Standard - 2.5.10; 3.2; 3.7; Biomass Geological Storage - 5.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation for injection and pre-processing sites.

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Stakeholder Input Process	Process for gaining input from entities and organizations with an interest in or impact from the project	Yes	Environmental and Socioeconomic Impacts	The Project defined and followed a Stakeholder Input Process which meets the requirements set out in the Standard	Standard - 3.5	Colorado Outreach and Engagement 2024.pdf V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed with recommended improvements. Evidence of reasonable community engagement via community support activities. However, limited open community outreach and engagement has occurred, such as open town hall meetings. These are currently planned for November / December 2024. Strongly recommend confirming results of these activities in conjunction with future verifications and monitor activities in future (continuing efforts)
Grievance Mechanism	Process for addressing concerns from entities and organizations with an interest in or impact from the project	Yes	Environmental and Socioeconomic Impacts	The Stakeholder Input Process include a Grievance Mechanism for stakeholders to voice, process and resolve grievances	Standard - 3.5	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed. Charm Ft Lupton site has listing with chamber of commerce, phone number available on site and Charm site, and a community engagement manager that inquiries are directed to. No changes to process at pre-process or injection site

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Site Safety Program	Site safety programs, especially as related to production and handling of biomass	Yes	Environmental and Socioeconomic Impacts	Documentation of safety programs and compliance	Bio-oil Geological Storage - 5.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx EHS-0006 Emergency Action Plan - MiniForge.pdf EPA ID.pdf EHS-0011 Chemical Hygiene Plan (CHP) & Hazard Communication Program (HCS) - Google Docs.pdf Monthly Site Inspections -Template.pdf EHS Training Modules - MiniForge - Sheet1.pdf ADP MyLearning for MRV - Operators.pdf Charm Bio-Oil SDS.pdf Monthly Site Inspections -Template.pdf EHS Training Modules - MiniForge - Sheet1.pdf ADP MyLearning for MRV - Operators.pdf Charm Bio-Oil SDS.pdf SOP-0011 Bighorn Standard Operating Procedure_REV-B (1).pdf SOP-0007 Biomass Screening System Standard Operating Procedures.pdf SOP-0001 Peanut Wagon.pdf MiniForge IBC Tote Bio-oil Sampling SOP.pdf WI-0901 Bio-Oil Transfer.pdf SOP-0001 Peanut Wagon.pdf MiniForge IBC Tote Bio-oil Sampling SOP.pdf WI-0901 Bio-Oil Transfer.pdf Reportable Safety Incidents TD.png	Confirmed. Charm has procedures, SOPs, training, and EHS requirements in place, managed by EHS mgr and site operations management
Legal ownership of Removals	Determination of the legal rights to Credits generated through CDR	Yes	Ownership	production and handling of biomass	Standard - 3.1; Bio-oil Geological Storage 6.3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation
Eligibility on Isometric Registry	Determination of the ability to list Credits generated through CDR on the Isometric Registry	Yes	Eligibility	The Project has confirmed that the activities of the Project are exclusively registered with the Isometric Registry, AND the Project uses the latest available version of a Certified Protocol (unless a grace period has been explicitly specified by Isometric), AND the Project meet all eligibility criteria defined by the selected Certified Protocol	Standard - 3.3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Confirmation via listing on Isometric Registry MRV portal	Confirmed

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Crediting Period	The period of time over which a Project Design Document is valid, and over which Removals may be Verified, resulting in Issued Credits	Yes	Crediting Period	The PDD specifies a Crediting Period which is no longer than the maximum period allowed by the Standard and the relevant Certified Protocol (5 years; unless otherwise specified)	Standard - 3.4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation . Crediting Period not changed (03/12/2024 - 5 yrs) - solely added Charm bio-oil production as eligible activity within the crediting period
Regulatory Compliance	Adherance to all jurisdictional requirements	Yes	Regulatory Compliance	The Project confirms in the PDD that it is in compliance with regulations for all jurisdictions to which the Project is beholden	Standard - 3.6	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed.
Regulatory Compliance - Injection Well Permitting	Maintains a valid UIC or equivalent permit	Yes	Regulatory Compliance	Project demonstrates a current valid UIC well permit valid for the injection period. Permit owner demonstrates full compliance with permit and provides supporting records, including required monitoring records, well design and installation records, and all compliance reports	Biomass Salt Caverns 1.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation
Regulatory Compliance - Injection Well Permitting	UIC Permit is Class V or equivalent	Yes	Regulatory Compliance	UIC Permit	Biomass Salt Caverns 1.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation
Regulatory Compliance - Injection Well Permitting	Permit specifically identifies biomass, bio-oil or equivalent type of injectate and covers emplaced material for validation / reporting period	Yes	Regulatory Compliance	UIC Permit	Biomass Salt Caverns 1.0	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Confirmed - no change from previous Validation Confirm compliance of Charm pyrolysis bio-oil with UIC permit requirements prior to injection and for future verifications

Biomass Feedstock Accounting

Parameter	Notes	Parameter Description	Required for Validation?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Applicability		Requirements are applicable to: Agricultural Residue Forestry Thinnings or byproducts of forestry operations Industrial biomass residues	Yes	All biomass feedstocks	Documentation of feedstock type and source		V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Biomass Logs.xlsx Weigh Ticket 041824.pdf (and others) 8_12 Day 0 Biomass Totals.xlsx Raw Wood Storage Site.xlsx Visual Observation at storage and processing sites Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Confirmed. Only feedstock used for pyrolysis at Ft Lupton is logs from forest thinning and fire prevention activities from Govt funded forest management project (altitude Forestry contractor)
Eligibility	**See eligbility criteria below: Select Claimed eligibility criteria ONLY**	**Identify Which category of eligibility the feedstock falls under: - Any of EC1-EC4 meets market leakage requirements OR - one of EC5-EC7 plus one of EC8-EC12 meets market leakage requirement**	Yes	All biomass feedstocks	See Below	2.1	See below	Feedstock meets criteria EC5, EC8, EC14, C1
Eligibility - Direct and Indirect Market Leakage Combined	EC1	NA				2.1 Table 2	NA	
Eligibility - Direct and Indirect Market Leakage Combined	EC2	NA				2.1 Table 2	NA	
Eligibility - Direct and Indirect Market Leakage Combined	EC3	NA				2.1 Table 2	NA	
Eligibility - Direct and Indirect Market Leakage Combined	EC4	NA				2.1 Table 2	NA	

Eligibility - Direct Market Leakage	EC5	Feedstock is a forest residue1.			The Project Proponent must demonstrate that the feedstock is an non-marketable wood product. For example, beetle kill, sticks and twigs, mill residues, etc.	2.1 Table 3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Biomass Logs.xlsx Weigh Ticket 041824.pdf (and others) 8_12 Day 0 Biomass Totals.xlsx Raw Wood Storage Site.xlsx Visual Observation at storage and processing sites Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Confirmed. Only feedstock used for pyrolysis at Ft Lupton is logs from forest thinning and fire prevention activities from Govt funded forest management project (altitude Forestry contractor)
Eligibility - Direct Market Leakage	EC6	NA			NA	2.1 Table 3	NA	
Eligibility - Direct Market Leakage	EC7	NA			NA	2.1 Table 3	NA	
Eligibility - Indirect Market Leakage	EC8	Applicable to forest residues or downstream wood wastes only: Biomass originates from a regulated forest management project.			The Project can provide a reference that the feedstock came from either: a NEPA, or state level equivalent2, planned project or a government regulated forest management project or a state or federal funded or managed forest management activity In each case, the project with the associated harvest plan will be subject to competent Verification & Validation Body to ensure adherence to the criteria.	2.1 Table 4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Biomass Logs.xlsx Weigh Ticket 041824.pdf (and others) 8_12 Day 0 Biomass Totals.xlsx Raw Wood Storage Site.xlsx Visual Observation at storage and processing sites Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Confirmed. Only feedstock used for pyrolysis at Ft Lupton is logs from forest thinning and fire prevention activities from Govt funded forest management project (altitude Forestry contractor)
Eligibility - Indirect Market Leakage	EC9	NA			NA	2.1 Table 4	NA	
Eligibility - Indirect Market Leakage	EC10	NA			NA	2.1 Table 4	NA	

Eligibility - Indirect Market Leakage	EC11	NA			NA	2.1 Table 4	NA	
Eligibility - Indirect Market Leakage	EC12	NA			NA	2.1 Table 4	NA	
Recommended Sourcing Principles	Not required, but recommended - document if appropriate	Historical land use: purchase feedstock only from acreages that have been used to grow corn either continuously or in rotation with another crop for the past 10 years	No		Farm Production Records for 10 yrs	2.1.1	NA	
Recommended Sourcing Principles	Not required, but recommended - document if appropriate	Tillage practices: for crop residues sourced from row crop agriculture, collect evidence from the feedstock supplier on the intensity of cultivation practices on the acreage from which the corn stover is sourced. Project Proponents are recommended to maintain records on whether fields engage in conventional tillage, reduced tillage, or no tillage.	No		Farm tillage practices records	2.1.1	NA	
Recommended Sourcing Principles	Not required, but recommended - document if appropriate	Sustainable feedstock harvest: collect evidence from the feedstock supplier that the rate of biomass harvest on the acreage from which the feedstock is sourced does not exceed the sustainable rate of removal	No		Farm production records and calculations of sustainable rate of removal	2.1.1	NA	
Counterfactual Storage Eligibility	EC13	NA			NA	2.2 Table 5	NA	

Counterfactual Storage Eligibility	EC14	Feedstocks that are sourced through government managed or funded wildfire mitigation or restoration activities will be assigned a counterfactual storage value of 0.			Project Proponent should provide a government source demonstrating that the funding or management of the biomass collection or harvesting was required, funded, or managed by a federal or state government body and was done for the purpose of wildfire mitigation or post-wildfire ecosystem restoration.		V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Biomass Logs.xlsx Weigh Ticket 041824.pdf (and others) 8_12 Day 0 Biomass Totals.xlsx Raw Wood Storage Site.xlsx Visual Observation at storage and processing sites Copy of Evidence Package- CO Woody Biomass.pdf Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf	Confirmed. Only feedstock used for pyrolysis at Ft Lupton is logs from forest thinning and fire prevention activities from Govt funded forest management project (altitude Forestry contractor)
Counterfactual Storage Accounting		Calculation of the stored biogenic C that is eligible for crediting under the protocol is calculated appropriately based on counterfactual eligibility criteria EC13 or EC14 as appropriate	Yes		Calculations of counterfactual storage and eligible stored biogenic CO2	2.2	NA	
Dedicated Energy Feedstock Eligibility	EC15	NA			NA		NA	

GHG Accounting

Parameter	Notes	Parameter Description	Required for Validation?	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
CO2 Counterfactual	See Biomass Feedstock Eligibility criteria for evidence requirements to support calculations	Biomass feedstock counterfactual (ineligible for crediting and deducted from LCA)	Yes	LCA calculations - inclusion of CO2 counterfactual	Biomass Feedstock Accounting 3.2	See Biomass Feedstock Accounting Tab	Counterfactual = 0 (C1 / EC 14 criteria)
CO2e Leakage / replacement	See Biomass Feedstock Eligibility criteria for evidence requirements to support calculations. See Table 7 Biomass Feedstock Accounting Module for exemption criteria for replacement emissions accounting: - Feedstock currently serves no purpose (or is replaced by one that serves no purpose) - feedstock usage is only for amounts above the sustainable use rate	Total emissions associated with market leakage for biomass feedstock. Must include: - GHG emissions associated with energy consumption for replacement activities - embodied emissions associated with replacement activities - transport emissions associated with replacement activities - environmental service replacement (i.e. fertilizer)	Yes	LCA calculations - inclusion of CO2 Leakage / Replacement AND -Affidavit from feedstock supplier documenting past use of their biomass over last 5 years -Historical evidence of feedstock supplier behavior in region demonstrating that the potential best use for the biomass feedstock is not applicable (i.e. it is not economical to truck it more than a specified distance) - documentation of sustainable use rate if exemption C3 used	Biomass Feedstock Accounting 3.2	See Biomass Feedstock Accounting Tab / NA	Leakage = 0 (fire prevention / government program)
System Boundary		GHG Statement accounts for all sources sinks and reservoirs (including embodied emissions and cradle to grave analysis) associated with the project, including but not limited to: biomass production biomass transport biomass conversion bio-oil transport and storage bio-oil injection monitoring	Yes	PDD documentation and GHG Statement demonstrating system boundary and inclusion of all required GHGs; Visual observation of process and confirmation of all inputs and outputs LCA calculations and report	Bio-oil Geologic Storage 7.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations.	GHG Statement and Calculation details meet requirements for LCA calculations and indicate likely Net negative LCA. Full LCA is not provided as this is for a validation update with LCA to be provided at time of future verification. GHG Statement outline and details provided document appropriate system boundary with all SSRs properly included
GHGs included		Assessment of GHG emissions in GHG Statement and LCA must account for all GHG emissions, as described in protocol (Table 1 Bio-oil Geologic Sequestration Section 7.1), including CO2, CH4, and N2O where required	Yes	PDD documentation and GHG Statement demonstrating system boundary and inclusion of all required GHGs; Visual observation of process and confirmation of all inputs and outputs LCA calculations and report	Bio-oil Geologic Storage 7.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations.	GHG Statement and Calculation details meet requirements for LCA calculations and indicate likely Net negative LCA. Full LCA is not provided as this is for a validation update with LCA to be provided at time of future verification.

Parameter	Notes	Parameter Description	Required for Validation?	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
							GHG Statement outline and details provided document appropriate system boundary with all GHGs accounted for in emission factors used (audited multiple factors to verify).
CO2e Removal Calculation	See 'Measurements - Net CO2 Calc' tab for all required measurements related to LCA	CO2e Removal calculation is calculated as CO2 e stored - CO2e Counterfactual - CO2e Emissions		PDD documentation and GHG Statement demonstrating system boundary and inclusion of all required GHGs; Visual observation of process and confirmation of all inputs and outputs LCA calculations and report	Bio-oil Geologic Storage 7.4	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations.	GHG Statement GHG Statement and Calculation details meet requirements for LCA calculations and indicate likely Net negative LCA. Full LCA is not provided as this is for a validation update with LCA to be provided at time of future verification. Calculation details meet requirements for determination of CO2eRemoval
CO2eStored Calculation	See 'Measurements - Net CO2 Calc' tab for all required measurements related to LCA	CO2e Stored is calculated as required ((Cbio-oil*Minj)/CCO2)		PDD documentation and GHG Statement demonstrating system boundary and inclusion of all required GHGs; Visual observation of process and confirmation of all inputs and outputs LCA calculations and report	Bio-oil Geologic Storage 7.4.1 (Eq3)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations.	GHG Statement GHG Statement and Calculation details meet requirements for LCA calculations and indicate likely Net negative LCA. Full LCA is not provided as this is for a validation update with LCA to be provided at time of future verification. Calculation details meet requirements for determination of CO2eStored. Bio-oil sampling and analysis for carbon content has not been initiated as of this validation (occurs just prior to processing for injection). Confirm analysis and utilization of results in calculating CO2eStored in future verification

Parameter	Notes	Parameter Description	Required for Validation?	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Spills, leaks, releases & process upsets		Any bio-oil spills, leaks or releases or process upsets which may impact GHG emissions accounting must documented and included in LCA as appropriate		logs of spills, leaks, process upsets LCA calculations & Report	Bio-oil Geologic Storage 7.4.1.3	BGH2 p1.jpg BGH2 p2.jpg Pump Log.jpg BGH1 p2.jpg BGH1 p1.jpg BGH2 p2.jpg Pump Log 1.jpg BGH2 p1.jpg Pump Log 2.jpg Digital Data 8_20 + 21.xlsx	Incidents are logged on operator logs. Logs are reviewed by operations management to confirm data and info. No incidents have been identified to date, so difficult to confirm that any such data is accounted for in MRV/LCA. To be confirmed in future verifications.

Parameter	Notes	Parameter Description	Required for Validation?	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Calculation of CO2eEmissions	See 'Measurements - Net CO2 Calc' tab for all required measurements related to LCA	CO2eEmissions is calculated in accordance with Eq 7 and requirements of Energy Emissions Module, Transportation Emissions Module, and Embodied Emissions Module and Misc. emissions requirements		PPD documentation and GHG Statement demonstrating system boundary and inclusion of all required GHGs; Visual observation of process and confirmation of all inputs and outputs LCA calculations and report	Bio-oil Geologic Storage 7.4.3 (Eq7)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations. AB~INVOICE_4GUT170424519.PDF Office Generator Refill Log.xlsx BGN Spare Parts and Consumable Estimates.xlsx Data Base -- Updated.xlsx Pyrolysis MRV Input and Output Measurements Summary.pptx Pyrolysis MRV at MiniForge.pptx Digital Data 8_20 + 21.xlsx ce00_9152482239_invoice_20240807_020331.pdf Charm 8145951006.pdf Charm 8145881078.pdf Charm 8145556570.pdf Charm 8145752481.pdf	GHG Statement GHG Statement and Calculation details meet requirements for LCA calculations for CO2eEmissions. Full LCA is not provided as this is for a validation update with LCA to be provided at time of future verification. Calculation details meet requirements for determination of CO2eStored. Note that Transportation emissions utilize "the average “deadhead” or unloaded journey distance for a vehicle of this type according to USDA analysis". This is sufficient for accounting for return trips, but it may be preferable to use return trip to logging site if the trip route is known. Also note that several process changes are occurring as of time of this validation (September, October, 2024), including change of pyrolysis tailgas flare to alternative burner, addition of more pyrolyzers, optimization of operating conditions, and other changes that must be accounted for in updated process emissions calculations for the LCA

Parameter	Notes	Parameter Description	Required for Validation?	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Calculation of CO2e Tailgas (bio-oil conversion specific)		Direct emissions from pyrolysis process (tailgas) are accounted for and measured in accordance with 7.4.3.6		PPD documentation and GHG Statement demonstrating system boundary and inclusion of all required GHGs; Visual observation of process and confirmation of all inputs and outputs LCA calculations and report Emissions Testing Report Tailgas flow meter data	Bio-oil Geologic Storage 7.4.3.5	V 1.7 Charm IndV 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD provides description of Charm Biooil sequestration GHG statement development and all calculations. Copy of Emissions Performance Test Report EM-24-1203-001 REV 0.pdf Copy of Emissions Calculations - Methane.xlsx	Tail gas emissions are appropriately calculated based on qualified stack test. Note that operating conditions on initial verifications using Charm pyrolysis oil should be confirmed and compared to stack test conditions. Note that Charm is planning to modify the flare system with a low swirl burner in September / October, 2024. And additional stack test will be completed on the new tailgas pollution control device in this period. This data must be used for future verifications and LCA calculations.

Measurements – Net CO2 Calculation

Parameter	Parameter Description	Required by the protocol	Equation	Parameter Type	Units	Data Source	Measurement Method	Monitoring Frequency	QA/QC Procedures	Required Evidence	Reference in Protocol	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
CBio–oil, n	%wt of C in the bio-oil injectant	Always	Eq. 4 (Bio-oil Geological Storage)	Measured	wt%	Analytical determination of carbon content of bio-oil	ASTM D5271, ASTM D5373, NREL or similar / equivalent method for carbon content analysis	Measure per injection/production batch, or if 30 samples with the same feedstock type & processing have been taken within a 6 month period, this may be changed to a minimum of every 10 production batches, as well as random sampling. Minimum number of 3 samples per Batch measured, unless minimal 'within batch' variation can be justified. See 'Biomass Carbon Content Measurement' section for full details	ISO 17025 accredited laboratory	Analytical reports from qualified laboratory for audited samples, including supporting lab QA/QC results	7.4.1 (Bio-oil Geological Storage)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx SOP-0011 Bighorn Standard Operating Procedure_REV-B (1).pdf MiniForge IBC Tote Bio-oil Sampling SOP.pdf Charm Verification Report 2024-07-26	Monitoring Plan is in place to include required sampling and analysis in compliance with protocol. Analysis has not been completed. Must be confirmed at verification
mInj, n*mInj, n*	Total mass of bio-oil injectant	Always	Eq. 4 (Bio-oil Geological Storage)	Measured	kg	Direct mass measurement	Calibrated Weigh Scale	Each bio-oil delivery	Scales must be calibrated annually by certified entity	Weigh scale tickets for each delivery of bio-oil (arrival and departure weights) or equivalent; Calibration records for scales	7.4.1 (Bio-oil Geological Storage)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	NA - no change from previous Validation. Confirm at future verification
CBio–oil, p*CBio–oil, p*	%wt of C in the bio-oil injectant	Under certain conditions:	Eq. 4 (Bio-oil Geological Storage)	Measured	wt%	Analytical determination of carbon content of bio-oil	ASTM D5291, NREL Laboratory Analysis Procedure for Determination of Carbon, Hydrogen, and Nitrogen in Bio-oils, or equivalent	One sample per production batch	ISO 17025 accredited laboratory	Analytical reports from qualified laboratory for audited samples, including supporting lab QA/QC results	7.4.1 (Bio-oil Geological Storage)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	NA - no change from previous Validation. Confirm at future verification
mInj, p*mInj, p*	Total mass of bio-oil injectant	Under certain conditions:	Eq. 4 (Bio-oil Geological Storage)	Measured	kg	Direct mass measurement	Calibrated Weigh Scale	Each Bio-oil delivery	Scales must be calibrated annually by certified entity	Weigh scale tickets for each delivery of bio-oil (arrival and departure weights) or equivalent; Calibration records for scales	7.4.1 (Bio-oil Geological Storage)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	NA - no change from previous Validation. Confirm at future verification

mTailgas*mTailgas*	Mass flow rate of tail gas from biomass pyrolysis	Always	Eq. 8 (Bio-oil Geological Storage)	Measured or calculated	kg/hr	Direct flow measurement Emissions testing results Material balance calculation	Volumetric or mass flow meter direct measurement Emissions testing flow data Calculation based on material balance (carbon content of biomass feed - carbon content of bio oil)	Semi-continuous (flow rate) Single emissions test for each distinct feedstock or operating condition set Material balance calculation per batch	Flow meters calibrated by ISO 17025 accredited metrology laboratory (upon purchase and in accordance with manufacturer specification) Emissions test data collected by Stack Testing Accreditation Council accredited company. See 11.6.3. Material balance based on C content analysis by accredited laboratory	Flow meter data from plant data historian Emissions testing report Material Balance Calculation spreadsheets and C content analytical lab reports	7.4.3.5 (Bio-oil Geological Storage)	Copy of Emissions Performance Test Report EM-24-1203-001 REV 0.pdf Copy of Emissions Calculations - Methane.xlsx V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Tail gas emissions are appropriately calculated based on qualified stack test. Note that operating conditions on initial verifications using Charm pyrolysis oil should be confirmed and compared to stack test conditions. Note that Charm is planning to modify the flare system with a low swirl burner in September / October, 2024. And additional stack test will be completed on the new tailgas pollution control device in this period. Thsis data must be used for future verifications and LCA calculations.
CTailgas, CO2*CTailgas, CO2*	Concentration of CO_2 in tailgas	Always	Eq. 8 (Bio-oil Geological Storage)	Measured	wt%	On-line analyzer Emissions testing results	Gas chromatography, NDIR analyzer or similar EPA Method 3A, 3B, 3C or equivalent or other approved US EPA or CARB test methods for CO_2 emissions	Continuous monitoring preferred. For emissions testing, once per set of unique feedstock and operating conditions	Analyzers calibrated by ISO 17025 accredited metrology laboratory (upon purchase and in accordance with manufacturer specification) and with NIST-traceable calibration gas with CH4 and CO_2 concentration within 30% of tailgas concentration. Emissions test data collected by Stack Testing Accreditation Council accredited company. See 11.6.3.	Analyzer data from plant data historian or other sources Emissions testing report	7.4.3.5 (Bio-oil Geological Storage)		
CTailgas, CH4*CTailgas, CH4*	Concentration of CH4 in tailgas	Always	Eq. 8 (Bio-oil Geological Storage)	Measured	wt%	On-line analyzer Emissions testing results	Gas chromatography, NDIR analyzer or similar EPA Method 18 or equivalent or other approved US EPA or CARB test methods for CH4 emissions	Continuous monitoring preferred. For emissions testing, once per set of unique feedstock and operating conditions	Analyzers calibrated by ISO 17025 accredited metrology laboratory (upon purchase and in accordance with manufacturer specification) and with NIST-traceable calibration gas with CH4 and CO_2 concentration within 30% of tailgas concentration. Emissions test data	Analyzer data from plant data historian or other sources Emissions testing report	7.4.3.5 (Bio-oil Geological Storage)		

									collected by Stack Testing Accreditation Council accredited company. See 11.6.3.				
tp*tp*	Biomass conversion batch operation time	Always	Eq. 8 (Bio-oil Geological Storage)	Measured	hr	Operator logs or automated plant data acquisition system / historian	Data historian time stamp or operator timing / stopwatch	Each batch	Review of batch duration and validation of batch production time	Operator logs or data from plant data historian	7.4.3.5 (Bio-oil Geological Storage)	BGH2 p1.jpg BGH2 p2.jpg Pump Log.jpg BGH1 p2.jpg BGH1 p1.jpg BGH2 p2.jpg Pump Log 1.jpg BGH2 p1.jpg Pump Log 2.jpg Digital Data 8_20 + 21.xlsx V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Manual operator logs are utilized to document all operations, including start time, burn-in time, and end time of pyrolysis operationsmanual logs are transferred to digital data files and data eventually uploaded to charm register
mfuel, conversio n*mfuel, conversio n*	Mass of fuel used in biomass conversion	Always	Eq. 5 (Energy Use Accounting Module)	Measured	gal	Fuel usage records	Fuel meters Fuel container weight Fuel purchases or utility bills Equipment hours of operation (handling equipment only)	Each batch	Appropriate calibration and maintenance of scales or meters	Operator logs, plant data systems, or plant records	7.4.3.2 (Bio-oil Geological Storage)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf AB~INVOICE_4GUT170424519.PDF Office Generator Refill Log.xlsx Pyrolysis MRV Input and Output Measurements Summary.pptx Pyrolysis MRV at MiniForge.pptx Digital Data 8_20 + 21.xlsx ce00_9152482239_invoice_20240807_020331.pdf (propane invoice) Charm 8145951006.pdf (propane delivery) Charm 8145881078.pdf Charm 8145556570.pdf Charm 8145752481.pdf	records of propane tank fills (sales) are kept as are diesel tank fill / purchases
EFFuel,Conversio n*EFFuel,Conversi on*	Fuel emission factor for biomass conversion	Always	Eq. 5 (Energy Use Accounting Module)	Estimated	CO_2e/unit (tonnes)	Argonne National Laboratory GREET Model, California Air Resources Board modified GREET model (CA-GREET), Ecoinvent database, US Federal Life Cycle Inventory database or LCA Commons, or from similar databases used in common LCA practices or tools	N/A	Each batch	N/A	Choice and rationale for EF choice	7.4.3.2 (Bio-oil Geological Storage); 3.3 (Energy Use Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Confirmed

kwhConversion*kwhConversion*	Electricity usage for biomass conversion	Always	Eq. 2 (Energy Use Accounting Module)	Measured	kwh	Electricity usage records	Electricity meters OR Utility bills OR Equipment time of use and power rating	Each batch	Appropriate calibration and maintenance of meters	Operator logs, plant data systems, or plant records	Section 7.4.3.2 (Bio-oil Geological Storage); 3.2 (Energy Use Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx. export (5).xls (electricty meter data export)	Confirmed. Electricity meter data is monitored and exported for each operating day.
EFElect,Conversion*EFElect,Conversion*	Electricity emission factor	Always	Eq. 2 (Energy Use Accounting Module)	Estimated	CO_2e/kwh (tonnes)	Argonne National Laboratory GREET Model, California Air Resources Board modified GREET model (CA-GREET), Ecoinvent database, US Federal Life Cycle Inventory database or LCA Commons, or from similar databases used in common LCA practices or tools	N/A	Each batch	N/A	Choice and rationale for EF choice	7.4.3.2 (Bio-oil Geological Storage); 3.2 (Energy Use Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Confirmed
FConversion*FConversion*	Quantity of fuel used in transport of biomass to conversion site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Measured or estimated	gal	Vehicle or fleet management records	Fuel flow meters, fleet management system data, vehicle on board diagnostics, or similar	All deliveries for a batch	Verify instrument calibrations as appropriate	Meter, management system, OBD or other data records or logs, shipping documents	7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Confirmed
FInjection*FInjection*	Quantity of fuel used in transport of bio-oil to injection site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Measured or estimated	gal	Vehicle or fleet management records	Fuel flow meters, fleet management system data, vehicle on board diagnostics, or similar	All deliveries for a batch	Verify instrument calibrations as appropriate	Meter, management system, OBD or other data records or logs, shipping documents	7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Biomass Logs.xlsx Weigh Ticket (multiple PDFs) 8_12 Day 0 Biomass Totals.xlsx Raw Wood Storage Site.xlsx	Confirmed

EFFuel,Transportation*EFFuel,Transportation*	Fuel emission factor for transportation	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Estimated	CO_2e/unit (tonnes)	Argonne National Laboratory GREET Model, California Air Resources Board modified GREET model (CA-GREET), Ecoinvent database, US Federal Life Cycle Inventory database or LCA Commons, or from similar databases used in common LCA practices or tools	N/A	All trips for each batch	N/A	Choice and rationale for EF choice	7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Confirmed
DConversion*DConversion*	Biomass transportation distance traveled - feedstock supplier to conversion site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Measured or estimated	mi or km	Shipping records (bill of lading) OR Fleet management records OR Weighscale tickets	On-line mapping systems using origin and departure from shipping documents, odometer readings	All trips for each batch	Review and check of shipping records and origin/destination	Shipping records	Section 7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Biomass Logs.xlsx Weigh Ticket (multiple PDFs) 8_12 Day 0 Biomass Totals.xlsx Raw Wood Storage Site.xlsx Screenshot 2024-08-28 at 9.04.43 PM (google Maps of route)	Confirmed
WConversion*WConversion*	Mass of biomass transported from supplier to conversion site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Measured	kg, tonne, lb	Shipping records (bill of lading) OR Fleet management records OR Weighscale tickets	Calibrated weigh scale	All deliveries for a batch	Review weigh scale calibration certificate	Shipping records, weigh scale ticket	Section 7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)		Confirmed
DInjection*DInjection*	Bio-oil transportation distance traveled - conversion site to injection site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Measured or estimated	mi or km	Shipping records (bill of lading) OR Fleet management records OR Weighscale tickets	On-line mapping systems using origin and departure from shipping documents, odometer readings	All trips for each batch	Review and check of shipping records and origin/destination	Shipping records	Section 7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)		Confirmed

WInjection*WInjection*	Mass of bio-oil transported from conversion site to injection site	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Measured	kg, tonne, lb	Shipping records (bill of lading) OR Fleet management records OR Weighscale tickets	Calibrated weigh scale	All deliveries for a batch	Review weigh scale calibration certificate	Shipping records, weigh scale ticket	Section 7.4.3.3 (Bio-oil Geological Storage); 3.2 (Transportation Emissions Accounting Module)		Confirmed
EFTransportation, j*EFTransportation,j*	The weight- and distance-based emission factor for transportation	Under certain conditions	Eq. 2 (Transportation Emissions Accounting Module)	Estimated	CO_2e/unit (tonnes)	Argonne National Laboratory GREET Model, California Air Resources Board modified GREET model (CA-GREET), Ecoinvent database, US Federal Life Cycle Inventory database or LCA Commons, or from similar databases used in common LCA practices or tools	N/A	All trips for each batch	N/A	Choice and rationale for EF choice	7.4.3.3 (Bio-oil Geological Storage); 3.3 (Transportation Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf	Confirmed
mfuel, Injection*mfuel, Injection*	Mass of fuel used in bio-oil injection	Always	Eq. 5 (Energy Use Accounting Module)	Measured	gal	Fuel usage records	Fuel meters Fuel container weight Fuel purchases or utility bills Equipment hours of operation (handling equipment only)	Each batch	Appropriate calibration and maintenance of scales or meters	Operator logs, plant data systems, or plant records	Section 7.4.3.2 (Bio-oil Geological Storage); 3.3 (Energy Use Accounting Module)	NA	NA - no change from previous Validation. Confirm at future verification
EFFuel,Injection*EFFuel,Injection*	Fuel emission factor for bio-oil injection	Always	Eq. 5 (Energy Use Accounting Module)	Estimated	CO_2e/unit (tonnes)	Argonne National Laboratory GREET Model, California Air Resources Board modified GREET model (CA-GREET), Ecoinvent database, US Federal Life Cycle Inventory database or LCA Commons, or from similar databases used in common LCA practices or tools	N/A	Each batch	N/A	Choice and rationale for EF choice	7.4.3.2 (Bio-oil Geological Storage); 3.3 (Energy Use Accounting Module)	NA	NA - no change from previous Validation. Confirm at future verification

kwhInjection*kwh Injection*	Electricity usage for bio-oil injection	Always	Eq. 2 (Energy Use Accounting Module)	Measured	kwh	Electricity usage records	Electricity meters OR Utility bills OR Equipment time of use and power rating	Each batch	Appropriate calibration and maintenance of meters	Operator logs, plant data systems, or plant records	Section 7.4.3.2 (Bio-oil Geological Storage); 3.2 (Energy Use Accounting Module)	NA	NA - no change from previous Validation. Confirm at future verification
EFElect,Injection*EFElect,Injection*	Electricity emission factor	Always	Eq. 2 (Energy Use Accounting Module)	Estimated	CO_2e/kwh (tonnes)	Argonne National Laboratory GREET Model, California Air Resources Board modified GREET model (CA-GREET), Ecoinvent database, US Federal Life Cycle Inventory database or LCA Commons, or from similar databases used in common LCA practices or tools	N/A	Each batch	N/A	Choice and rationale for EF choice	7.4.3.2 (Bio-oil Geological Storage); 3.2 (Energy Use Accounting Module)	NA	NA - no change from previous Validation. Confirm at future verification
Product Stage Emissions	Includes raw material sourcing, transport to facility and manufacturing	Always		Measured	tonnes	Independently verified LCAs for the material or product completed; an environmental product declaration (EPD) for a material or product completed and independently verified	Number/weight of each product or material used in the project facility and a corresponding EPD-based embodied carbon emission factor, OR emission factors from LCA life cycle databases, including USLCI database, Ecoinvent, ICE Database, and other published and peer-reviewed databases of embodied emissions factors and the number or weight (depending on emission factor units) of each product or material at the facility, OR overall total cost of equipment and facilities for the project and cost based embodied emission factors	Each site	ISO 14040 or similar guidelines; ISO 14025, ISO 21930, EN 15804 or equivalent standards including product EPDs as well as industry-wide EPDs	Operator logs, plant data systems, or plant records	7.4.3.4 (Bio-oil Geological Storage); 3.0 & 3.2 (Embodied Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Data Base -- Updated.xlsx BGN Spare Parts and Consumable Estimates.xlsx	Charm maintains a database of all equipment, spare parts, and consumables. All associated embodied emissions are accounted for in LCA using appropriate factors. Confirm incorporation of embodied emissions in LCA at future verifications.

Construction Stage Emissions	Includes transport to site and installation at site	Always		Measured	tonnes	Independently verified LCAs for the material or product completed; or an environmental product declaration (EPD) for a material or product completed and independently verified	Number/weight of each product or material used in the project facility and a corresponding EPD-based embodied carbon emission factor, OR emission factors from LCA life cycle databases, including USLCI database, Ecoinvent, ICE Database, and other published and peer-reviewed databases of embodied emissions factors and the number or weight (depending on emission factor units) of each product or material at the facility, OR overall total cost of equipment and facilities for the project and cost based embodied emission factors	Each site	ISO 14040 or similar guidelines; ISO 14025, ISO 21930, EN 15804 or equivalent standards including product EPDs as well as industry-wide EPDs	Operator logs, plant data systems, or plant records	7.4.3.4 (Bio-oil Geological Storage); 3.0 & 3.2 (Embodied Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Data Base -- Updated.xlsx BGN Spare Parts and Consumable Estimates.xlsx	Charm maintains a database of all equipment, spare parts, and consumables. All associated embodied emissions are accounted for in LCA using appropriate factors. Confirm incorporation of embodied emissions in LCA at future verifications.
End of Life Stage Emissions	Includes demolition of building, transport to end of life, waste processing and final disposal or scenarios for these life cycle stages	Always		Measured	tonnes	Independently verified LCAs for the material or product completed; an environmental product declaration (EPD) for a material or product completed and independently verified	Number/weight of each product or material used in the project facility and a corresponding EPD-based embodied carbon emission factor, OR emission factors from LCA life cycle databases, including USLCI database, Ecoinvent, ICE Database, and other published and peer-reviewed databases of embodied emissions factors and the number or weight (depending on emission factor units) of each product or material at the facility, OR overall total cost of equipment and facilities for the project and cost based embodied emission factors	Each site	ISO 14040 or similar guidelines; ISO 14025, ISO 21930, EN 15804 or equivalent standards including product EPDs as well as industry-wide EPDs	Operator logs, plant data systems, or plant records	7.4.3.4 (Bio-oil Geological Storage); 3.0 & 3.2 (Embodied Emissions Accounting Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf Data Base -- Updated.xlsx BGN Spare Parts and Consumable Estimates.xlsx	Charm maintains a database of all equipment, spare parts, and consumables. All associated embodied emissions are accounted for in LCA using appropriate factors. Confirm incorporation of embodied emissions in LCA at future verifications.

Storage and Monitoring Emissions	The total quantity of GHG emissions associated with storage monitoring operations allocated to a removal	Always		Measured	tonnes	Electricity and fuel usage records; independently verified LCAs for the material or product completed; an environmental product declaration (EPD) for a material or product completed and independently verified	Electricity meters OR utility bills OR equipment time of use and power rating; fuel meters fuel container weight fuel purchases or utility bills equipment hours of operation (handling equipment only)	Each site	Appropriate calibration and maintenance of scales or meters	Operator logs, plant data systems, or plant records	7.4.3.4 (Bio-oil Geological Storage); 3.4 of applicable Storage Modules	NA	NA - no change from previous Validation. Confirm at future verification

Salt Cavern Requirements

Parameter	Parameter Description	Required?	Applicability	Required Evidence	Reference in Protocol or Standard	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
Storage Applicability / Eligibility	Bio-oil is injected into natural or engineered geologic formations for long duration storage by underground injection well	yes	Eligibility	Documentation of geologic storage formation and injection well permit	4.0 (bio-oil Geologic Storage)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Regulatory Compliance - Injection Well Permitting	Maintains a valid UIC or equivalent permit	Yes	Regulatory Compliance	Project demonstrates a current valid UIC well permit valid for the injection period. Permit owner demonstrates full compliance with permit and provides supporting records, including required monitoring records, well design and installation records, and all compliance reports	1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Regulatory Compliance - Injection Well Permitting	UIC Permit is Class V or equivalent	Yes	Permitting & Cavern Characterization	Class V permit	2.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Regulatory Compliance - Injection Well Permitting	Permit specifically identifies biomass, bio-oil or equivalent type of injectate and covers emplaced material for validation / reporting period	Yes	Permitting & Cavern Characterization	UIC Class V permit - language stating allowable emplacement materials to include biomass, etc.	2.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Cavern Characterization - Volume	Cavern is of sufficient volume to receive total anticipated volume of injectate stream and displaced brines (or those are otherwise properly disposed) and any biogas generated	Yes	Permitting & Cavern Characterization	UIC permit application or review report including studies or calculations documenting cavern volume and proposed injectate volume	2.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Cavern Characterization - Integrity	Cavern is free of transmissive faults or fractures	Yes	Permitting & Cavern Characterization	UIC permit application or review report including studies or of cavern structure, stability, and structural integrity	2.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Cavern Characterization - Integrity	Cavern has structural integrity (adequate wall thickness, not in close proximity to other caverns or salt boundary), no salt creep, and sufficient compressive and shear strength to prevent migration	Yes	Permitting & Cavern Characterization	UIC permit application or review report including studies or of cavern structure, stability, and structural integrity	2.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Regulatory Compliance - Injection Well Permitting	Cavern formation area of Review (AOR) has been defined	Yes	Regulatory Compliance	UIC permit application or review report	2.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Cavern Characterization - Integrity	Cavern characterizaation is reviewed every five years or when site conditions warrant, with comparison of pre-emplacement	Yes	Permitting & Cavern Characterization	UIC permit application or review report	2.2, 3.1.2	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation

	project assumptions to actual measured conditions and updates as needed						
Well Construction	Construction in compliance with UIC permit	Yes	Regulatory Compliance	UIC permit well construction records and report	2.3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Well Construction	Any legacy wells within AOR are identified evaluated, and plugged if a risk to durability is identified	Yes	Regulatory Compliance	UIC permit application or review report, including documentation of AOR, assessment of legacy wells, documentation of well plugging	2.3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Well Construction	well materials compatible with fluids / injectate and meet API, ASTM, ISO or comparable standards as applicable (i.e. corrosion resistance)	Yes	Regulatory Compliance	PDD outlines the applicable material standards clearly, evidence is provided that such materials were used based on well construction records	2.3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Operations and Injections	Monitoring is completed in accordance with UIC permit and all required parameters are in compliance with permit	Yes	Regulatory Compliance	Compliance monitoring reports submitted to permitting authority for reporting period and / or additional data, lab reports, etc. as required to demonstrate compliance for periods not reported yet	3.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Injectate Sampling	For new feedstocks - sample of each emplacement batch are collected and well mixed and representative	Yes	Registry Protocol Compliance, LCA calculations		3.1.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation Re-confirm proper sampling is completed for charm pyrolysis bio-oil in future verifications
Injectate Sampling	For ongoing feedstock types - samples are collected regularly and analyzed to ensure variation in feedstock can be identified	Yes	Registry Protocol Compliance, LCA calculations		3.1.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation Re-confirm proper sampling is completed for charm pyrolysis bio-oil in future verifications
Injectate Sampling	For blended feedstock, samples are collected for each emplacement batch	Yes	Registry Protocol Compliance, LCA calculations		3.1.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation Re-confirm proper sampling is completed for charm pyrolysis bio-oil in future verifications
Well head gas monitoring and safety systems	Wellhead is monitored for gas consituents and includes automatic shutoffs and alarms if gas concetrations are detected above baselines and if system pressure exceeds permit limits	Yes	Regulatory Compliance, Registry Protocol Compliance, LCA calculations		3.1.3	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation Note that gas monitoring instrumentation is being upgraded at injection site and should be confirmed and documented in future verifications

Measurements – Durability in Salt Caverns

Parameter	Parameter Description	Monitoring phase	Required by the protocol	Measurement Method	Monitoring Frequency	QA/QC Procedures	Required Evidence	Reference in Module	Evidence, Documents, or Files Reviewed	Verifier Findings and Notes
CBiomass, n	%wt of C in the biomass injectant	Operation	Yes	ASTM D5373, NREL Laboratory Analysis Procedure for Determination of Carbon and Hydrogen in Biomass, or equivalent	Minimum of 3 samples every 10 trucks of the same biomass type	ISO 17025 accredited laboratory	Data logs/Data Acquisition System Output	3.1.1	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Monitoring Plan is in place to include required sampling and analysis in compliance with protocol. Analysis has not been completed. Must be confirmed at verification
pH	pH of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Yes	pH meter	Once	As per manufacturer calibration procedure	Analytical reports from qualified laboratory for audited samples, including supporting lab QA/QC results	1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Temperature	Temperature of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Yes	Temperature probe/sensor	Once	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Conductivity or other salinity measurement	Conductivity of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Yes	ASTM Designation D1125-82	Once	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
TOC of the formation water	TOC of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Yes	GC	Once	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
pH	pH of injection slurry	Operation	Yes	pH meter	One sample per day of injection	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Monitoring Plan is in place to include required sampling and analysis in compliance with protocol. Analysis has not been completed. Must be confirmed at verification
Temperature	Temperature of injection slurry	Operation	Yes	Temperature probe/sensor	One sample per day of injection	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.1 (Biomass or Bio-oil Storage in Salt Caverns Module)		
Conductivity or other salinity measurement	Conductivity/Chloride content or other alternative determination as required by the regulator of the injection slurry	Operation	Yes		One sample per day of injection	As per manufacturer calibration procedure	Analytical reports from qualified laboratory for audited samples, including supporting lab QA/QC results	3.1.1 (Biomass or Bio-oil Storage in Salt Caverns Module)		

δ13δ13 Carbon signature	δ13δ13 Carbon signature of the compounds of the biomass	Operation	Not required but helpful	IRMS or equivalent	One sample per feedstock type	ISO 17025 accredited laboratory or as per manufacturer calibration procedure	Permit, Testing Data	3.1.1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Internal Well Integrity (Corrosion monitoring)	Demonstration of internal mechanical integrity	Operation & Post Injection	Yes	As per permit requirements			Permit, Testing Data	3.1.2, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
External Well integrity	Demonstration of external mechanical integrity	Operation & Post Injection	Yes	As per permit requirements			Data logs/Data Acquisition System Output	3.1.2, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Temperature	Well bottom (Cavern) Temperature	Post Injection	Yes	Temperature sensor or logger	6 monthly	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Pressure	Well bottom (Cavern) Pressure	Operation & Post Injection	Yes	Pressure sensor	Continuous (O), daily (PI)	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Sonar Survey	Sonar surveys to determine cavern volume and fill	Operation & Post Injection	Yes	Sonar as per permit	Every 5 years, and when the cavern is at 50% and full capacity.		Permit, Testing Data	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Cavern Fill	Determination of fill depth and its evolution	Pre Injection	Yes	e.g., depth to top of emplaced material, cavern bottom sounding, tagging top of fill, wirelines	As per permit		Data logs/Data Acquisition System Output	3.1.1, 3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation

Total carbon content	%wt of C in the displaced brine	Operation	Yes	e.g., TOC analyzer	Measurement shall occur every two weeks for 3 months, if the measurements are consistent this can be changed to every month for 6 months, then quarterly for 2 years, and finally every 6 months if these are still consistent	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Temperature	Temperature of the displaced brine	Operation	Yes	Temperature sensor	Measurements should initially each injection batch, however frequency can be reduced once consistency between samples is statistically proven	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	Monitoring Plan is in place to include required sampling and analysis in compliance with protocol. Analysis has not been completed. Must be confirmed at verification.
pH	pH of the displaced brine	Operation	Yes	pH meter	Measurements should initially each injection batch, however frequency can be reduced once consistency between samples is statistically proven	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)		
Conductivity or other salinity measurement	Chloride content/conductivity of displaced brine	Operation	Yes	ASTM Designation D1125-82	Measurements should initially each injection batch,	As per manufacturer calibration procedure		3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)		

					however frequency can be reduced once consistency between samples is statistically proven					
Biomass constituents	Biomass constituents e.g., water, oxygen and nitrogen	Operation	Not required but helpful		One sample per feedstock type		Data logs/Data Acquisition System Output	3.1.3 (Biomass or Bio-oil Storage in Salt Caverns Module)		
Gas composition	Gas composition analysis CO_2, N, O_2, CH4, or VOC levels in well headspace with resolution of at least 0.01%vol	Operation & Post Injection	Under certain conditions: When gas is detected and measurable and not a result of any initial gas release from pre-existing gas stores within the cavern	Wellhead gas sampling; gas monitor if limits breached	Monthly (O), as specified in permit (PI)	As per manufacturer calibration procedure	Data logs/Data Acquisition System Output	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation Note that injection site is installing new gas monitoring instrumentation. Provide specifications and information for future verification
Surface Subsidence	Surface Subsidence	Operation & Post Injection	Yes	e.g., Insar/CGPS/GPS, reference point	Every 2 years		Data logs/Data Acquisition System Output	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Surface topography or determination of reference point	Surface topography or reference set point as a baseline for subsidence monitoring	Pre Injection	Yes	e.g., Insar/CGPS/GPS, or setting a reference point			Analytical reports from qualified laboratory for audited samples, including supporting lab QA/QC results	1 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
USDW quality	Monitoring of USDWs for increases in pressure and changes to chloride concentrations at a minimum	Operation & Post Injection	Required as per permit requirements, helpful	As per permit requirements			Analytical reports from qualified laboratory for audited samples, including supporting lab QA/QC results	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation

Gas isotope stability	Stable isotope compositions of carbon containing species - usually measure by IRMS	Operation & Post Injection	Under certain conditions: When gas is detected and measurable and not a result of any initial gas release from pre-existing gas stores within the cavern	IRMS/ Cavity ring down mass spectrometry		ISO 17025 accredited laboratory or as per manufacturer calibration procedure	Permit, Testing Data	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Cavern volume	Cavern volume prior to injection	Pre Injection	Yes	As per permit requirements			Permit, Testing Data	1, 2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Permeability	Permeability of the salt	Pre Injection	Yes	Either measured, historical or literature data			Permit, Testing Data	1, 2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Porosity & permeability	Porosity & Permeability of caprock and any interbeds	Pre Injection	Yes	Either measured, historical or literature data			Permit, Testing Data, historical data	1, 2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation
Subsurface features	Investigate pre-existing faults and presence of any interbeds	Pre Injection	Yes	Either measured (e.g., well logs), historical or literature data			Permit, Testing Data, historical data	1, 2 (Biomass or Bio-oil Storage in Salt Caverns Module)	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx	NA - no change from previous Validation

APPENDIX 2: SITE VISIT PHOTOS

Figure A2-1. Bio-oil product fraction storage at Charm MiniForge site

Figure A2-2. Biomass feedstock storage and chipper at Charm MiniForge offsite stockpile site

Figure A2-3. Existing flare for pyrolysis gas destruction at Charm MiniForge site

APPENDIX 3: DOCUMENTS SUBMITTED FOR REVIEW

Biomass Throughput	Biomass Logs.xlsx
Biomass Transport Emissions	Weigh Ticket 041824.pdf
Biomass Transport Emissions	Weigh Ticket 41124.pdf
Biomass Transport Emissions	Weigh Ticket 042924.pdf
Biomass Transport Emissions	Weigh Ticket 042324.pdf
Biomass Transport Emissions	Weigh Ticket 043024.pdf
Biomass Transport Emissions	Weigh Ticket 042524.pdf
Biomass Transport Emissions	Weigh Ticket 041724.pdf
Biomass Transport Emissions	Weigh Ticket 041224.pdf
Biomass Transport Emissions	Weigh Ticket 042424.pdf
Biomass Transport Emissions	Weigh Ticket 042624.pdf
Biomass Transport Emissions	Weigh Ticket 050124.pdf
Biomass Transport Emissions	Weigh Ticket 051824.pdf
Biomass Transport Emissions	Weigh Ticket 052424.pdf
Biomass Transport Emissions	Weigh Ticket 051024.pdf
Biomass Transport Emissions	Weigh Ticket 050724.pdf
Biomass Transport Emissions	Weigh Ticket 050924.pdf
Biomass Transport Emissions	Weigh Ticket 052624.pdf
Biomass Transport Emissions	Weigh Ticket 051724.pdf
Biomass Transport Emissions	Weigh Ticket 052824.pdf
Biomass Transport Emissions	Weigh Ticket 052524.pdf
Biomass Transport Emissions	Weigh Ticket 051424.pdf
Biomass Transport Emissions	Weigh Ticket 052024.pdf
Biomass Transport Emissions	Weigh Ticket 052124(1).pdf
Biomass Transport Emissions	Weigh Ticket 051624.pdf
Biomass Transport Emissions	Weigh Ticket 060224.pdf
Biomass Transport Emissions	Weigh Ticket 051524.pdf
Biomass Transport Emissions	Weigh Ticket 060424.pdf
Biomass Transport Emissions	Weigh Ticket 052824-2.pdf
Biomass Transport Emissions	Weigh Ticket 052724.pdf
Biomass Transport Emissions	Weigh Ticket 052124.pdf
Biomass Transport Emissions	Weigh Ticket 052924.pdf
Biomass Transport Emissions	Weigh Ticket 052224.pdf
Biomass Transport Emissions	Weigh Ticket 052324.pdf
Biomass Transport Emissions	Weigh Ticket 053024-2.pdf
Biomass Transport Emissions	Weigh Ticket 053024.pdf
Biomass Transport Emissions	Weigh Ticket 060424-2.pdf
Biomass Transport Emissions	Weigh Ticket 060824-2.pdf
Biomass Transport Emissions	Weigh Ticket 060924.pdf

Biomass Transport Emissions	Weigh Ticket 061524.pdf
Biomass Transport Emissions	Weigh Ticket 061624.pdf
Biomass Transport Emissions	Weigh Ticket 061724.pdf
Biomass Transport Emissions	Weigh Ticket 061424.pdf
Biomass Transport Emissions	weigh ticket 070224.pdf
Biomass Transport Emissions	Weigh Ticket 061324.pdf
Biomass Transport Emissions	Weigh Ticket 061024.pdf
Biomass Transport Emissions	Weigh Ticket 061024-2.pdf
Biomass Transport Emissions	Weigh Ticket 060824.pdf
Biomass Transport Emissions	Weigh Ticket 061124.pdf
Biomass Transport Emissions	Weigh Ticket 061224.pdf
Biomass Transport Emissions	Weigh Ticket 072124.pdf
Biomass Transport Emissions	Weigh Ticket 071024.pdf
Biomass Transport Emissions	weigh ticket 070524.pdf
Biomass Transport Emissions	weigh ticket 070824.pdf
Biomass Transport Emissions	Weigh Ticket 072524.pdf
Biomass Transport Emissions	Weigh Ticket 071824-2.pdf
Biomass Transport Emissions	Weigh Ticket 071324.pdf
Biomass Transport Emissions	Weigh Ticket 072424.pdf
Biomass Transport Emissions	weigh ticket 070424.pdf
Biomass Transport Emissions	weigh ticket 070324.pdf
Biomass Transport Emissions	Weigh Ticket 071524.pdf
Biomass Transport Emissions	Weigh Ticket 071824.pdf
Biomass Transport Emissions	Weigh Ticket 071124.pdf
Biomass Transport Emissions	Weigh Ticket 072024.pdf
Biomass Transport Emissions	weigh ticket 070524-2.pdf
Biomass Transport Emissions	Weigh Ticket 061824.pdf
Biomass Transport Emissions	Weigh Ticket 071624.pdf
Biomass Transport Emissions	weigh ticket 061924.pdf
Biomass Transport Emissions	Weigh Ticket 071424.pdf
Biomass Transport Emissions	Weigh Ticket 070924.pdf
Biomass Transport Emissions	weigh ticket 070624.pdf
Biomass Transport Emissions	Weigh Ticket 071224.pdf
Biomass Transport Emissions	8_12 Day 0 Biomass Totals.xlsx
Biomass Transport Emissions	Raw Wood Storage Site.xlsx
Decks	Pyrolysis MRV Input and Output Measurements Summary.pptx
Decks	Pyrolysis MRV at MiniForge.pptx
Diesel	AB~INVOICE_4GUT170424519.PDF
Diesel	Office Generator Refill Log.xlsx
Equipment Lists	BGN Spare Parts and Consumable Estimates.xlsx
Equipment Lists	Data Base -- Updated.xlsx

Example Sensor Datasets	BGN1_HotflowData_08212024.xlsx
Example Sensor Datasets	BGN2_HotflowData_08202024.xlsx
Example Sensor Datasets	dataExport-BGN3-Flare-08222024 (1).xlsx
Feedstock	Copy of Evidence Package- CO Woody Biomass.pdf
Feedstock	Copy of Charm Industrial - Altitude Forestry - Supply Chain Agreement FINAL .pdf
Feedstock	Copy of Weigh Ticket 071224.pdf
Operating Logs	BGH2 p1.jpg
Operating Logs	BGH2 p2.jpg
Operating Logs	Pump Log.jpg
Operating Logs	BGH1 p2.jpg
Operating Logs	BGH1 p1.jpg
Operating Logs	BGH2 p2.jpg
Operating Logs	Pump Log 1.jpg
Operating Logs	BGH2 p1.jpg
Operating Logs	Pump Log 2.jpg
Paper Sheets 8-20-21	Digital Data 8_20 + 21.xlsx
PDD	PDD Change-Log.xlsx
PDD	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf
People + Community Engagement	Colorado Outreach and Engagement 2024.pdf
People + Community Engagement	MiniForge Compensation Summary.docx
Propane	ce00_9152482239_invoice_20240807_020331.pdf
Propane	Charm 8145951006.pdf
Propane	Charm 8145881078.pdf
Propane	Charm 8145556570.pdf
Propane	Charm 8145752481.pdf
Safety + EHS Docs	Permit_24FTL-PW00057 - Ponds and Utilities
Safety + EHS Docs	EHS-0006 Emergency Action Plan - MiniForge.pdf
Safety + EHS Docs	EPA ID.pdf
Safety + EHS Docs	EHS-0011 Chemical Hygiene Plan (CHP) & Hazard Communication Program (HCS) - Google Docs.pdf
Safety + EHS Docs	Monthly Site Inspections -Template.pdf
Safety + EHS Docs	EHS Training Modules - MiniForge - Sheet1.pdf
Safety + EHS Docs	ADP MyLearning for MRV - Operators.pdf
Safety + EHS Docs	Charm Bio-Oil SDS.pdf
Safety + EHS Docs	Reportable Safety Incidents TD.png
SOPs and Work Instructions	SOP-0011 Bighorn Standard Operating Procedure_REV-B (1).pdf
SOPs and Work Instructions	SOP-0007 Biomass Screening System Standard Operating Procedures.pdf
SOPs and Work Instructions	SOP-0001 Peanut Wagon.pdf
SOPs and Work Instructions	MiniForge IBC Tote Bio-oil Sampling SOP.pdf
SOPs and Work Instructions	WI-0901 Bio-Oil Transfer.pdf
Tailgas+Air Permitting	Charm Industrial_Wood Chip Pyrolyzer Burner APEN Cover Letter (1).pdf
Tailgas+Air Permitting	Copy of Emissions Performance Test Report EM-24-1203-001 REV 0.pdf

Tailgas+Air Permitting	Air Permitting Pathway in CO.docx
Tailgas+Air Permitting	Copy of Emissions Calculations - Methane.xlsx
Validation Documents	Validation Visit Cast of Characters.docx
Validation Documents	MiniForge Inputs and Outputs Summary + Critical Measurements.xlsx
Validation Documents	Charm Verification Report 2024-07-26
Validation Documents	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1

APPENDIX 4: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	Criteria Met?	Evidence / Notes *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Kyann Hadife
350Solutions, Carbon Removal Verification Engineer

EDUCATION
Columbia University - MS Carbon Management (4.03/4.0 GPA)
University College London - BS Civil & Environmental Engineering (4.0/4.0 GPA)

SUMMARY
Kyann is an MS in Carbon Management graduate from Columbia University with experience in LCA/TEA consulting, laboratory analytical work, and engineering design. Prior to Columbia, he earned a bachelor's degree in civil and environmental Engineering from University College London. At UCL and Columbia Kyann has researched anaerobic digestion, Direct Air Capture and CO2 mineralization, developing prototypes, SOPs, and data analysis models. During his master's, Kyann also consulted for companies across hybrid and engineered CDR pathways including BiCRS, OAE, and Mineralization, on developing LCA/TEAs and site-selection tools, conducting market research, and writing applications for grants from the DOE, XPRIZE foundation, Remove and others. Most recently, Kyann has spearheaded Carbon Lockdown's Puro.earth verification for their FOAK terrestrial biomass burial project. Before being a full-time team member, Kyann worked as a research intern at 350Solutions in the summer of 2023.

RESEARCH AND PROFESSIONAL EXPERIENCE:
Mar 2024 – Present: 350Solutions, Carbon Removal Verification Engineer
At 350Solutions, Kyann has Verified CDR technologies on behalf of the XPRIZE Carbon Removal, registries challenge across pathways, including DAC, DOC, and mineralization. During his time at the company, he has also verified monitoring approaches (ISO14034) and validated LCA models (ISO14040) for carbon removal approaches.

Nov 2023 – Mar 2024: Carbon Lockdown Project, MRV Consultant
Kyann has spearheaded certification efforts for Carbon Lockdown's 5kt/y Potomac site under Puro.earth's Terrestrial Biomass Burial Methodology, co-developing the company's LCA, TEA, EIA, and MRV plan. He has also evaluated the eligibility of the project biomass under various sustainable biomass sourcing standards. As part of this role, Kyann participated in webinars and working groups organized by the DOE and national laboratories aiming to develop a unified LCA framework for terrestrial biomass burial.

Sep 2023 – Mar 2024: Brineworks, LCA/TEA Consultant
During his time at Brineworks, Kyann has conducted mass, energy, and carbon balances for 6+ carbon-negative brine treatment and material recovery pathways. He used LCAs and TEAs to inform process optimization efforts and achieve net negative process emissions while maintaining competitive brine treatment prices. He has also investigated the additionality and economics of a variety of CDR pathways unlocked by electrochemical brine treatment.

Sep 2023 – Dec 2023: Master's Thesis – Columbia University & Octavia Carbon
Conducted a cradle-to-gate LCA/TEA on project Hummingbird, the global south's first DAC + Mineralization plant. As part of the study, Kyann leveraged Aspen HYSYS and Adsorption models, along with statistical methods, to predict plant performance at scale based on values obtained from lab-scale testing. As part of his academic coursework at UCL and Columbia, Kyann has also conducted LCAs on various processes and products including Kenaf fiber insulation board, Steam Methane Reforming with pressure swing adsorption, and Biogenic/Algal CO2 conversion.

350Solutions, Inc. Corporate Experience

350Solutions serves as an independent expert in cleantech, low carbon, and environmental technologies. We provide an unbiased assessment of innovative technologies. 350Solutions is accredited through ANAB under ISO 17020 as an independent inspection body to provide independent technology evaluation services using the ISO 14034 ETV process. In addition, 350Solutions staff include a Certified Measurement and Verification Professional (CMVP for IPMVP) and a North Carolina Registered Professional Engineer (P.E.).